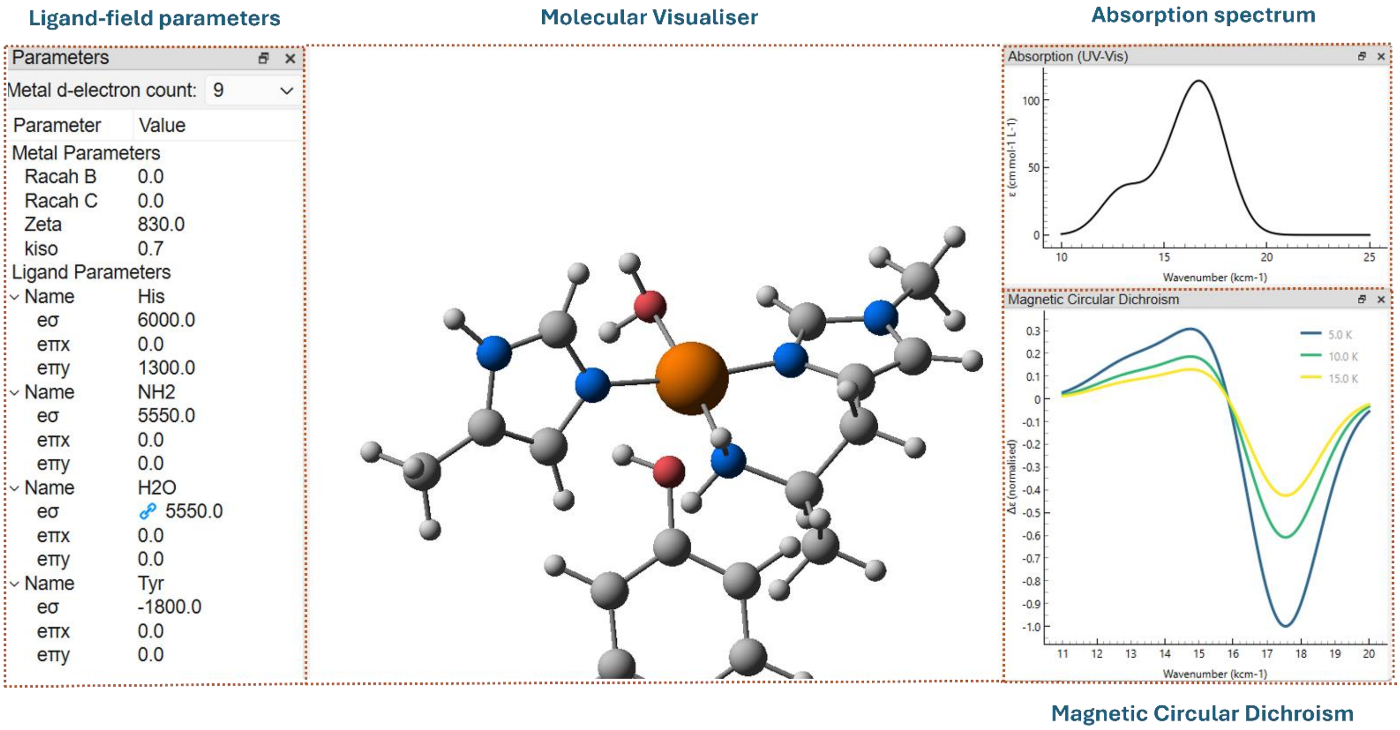
KESTREL User Manual



Version 1.0

chem-kestrel@york.ac.uk

https://chem-kestrel.github.io/

Contributors:

- George Nunn – codebase and GUI development
- Benjamin S. Grace – GUI development and hyperfine module
- Paul H. Walton
- Martin A. Bates

KESTREL is a computational package that performs ligand field calculations on any monomeric 3d transition metal complex in any coordination geometry. Based on the Angular Overlap Model (AOM) / Cellular Ligand Field (CLF) developed by Malcolm Gerloch[1], KESTREL assumes minimal spatial overlap between the metal and ligand orbitals and only calculates the energies of the metal d-orbitals. The effect of the ligands is then described using chemically significant parameter values, allowing calculations to be carried out and interpreted in real-time.

KESTREL will calculate d-d energies, magnetic properties such as Electron Paramagnetic Resonance g factors, metal hyperfine values and paramagnetic susceptibilities. The intensities of d-d transitions can also be calculated, permitting the simulation of absorption, circular dichroism (CD) and magnetic circular dichroism (MCD) spectra.

The combination of real-time calculations and chemically meaningful parameter values provides a powerful tool for relating the chemistry of a transition metal complex to its spectroscopic results. The following guide contains an "installation" guide for new users (pages 4-5) followed by detailed documentation on each feature and calculation. Finally, there is a FAQ section from page 32 to help with troubleshooting.

Due to the assumption of minimal spatial overlap, KESTREL is best suited to 3d transition metal complexes with ligands that do exhibit large covalent overlap. Extension to other systems is possible (i.e. 4d transition metals, or CO ligands), but the accuracy such calculations is uncertain.

Contents

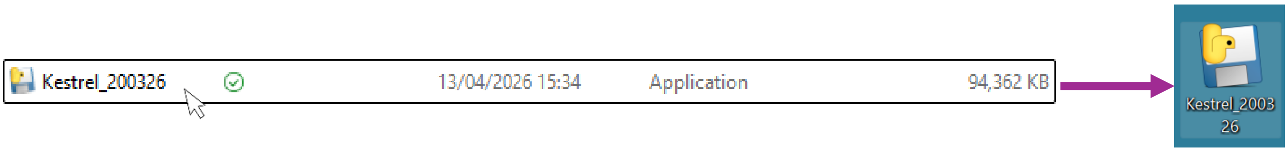
Installation

Before running any calculations in KESTREL, the graphical user interface (GUI) or python codebase needs to be installed. This guide will focus on the installation and use of the GUI.

KESTREL will run on computers that support Windows 10 or 11, Linux or MacOS 13 (ventura) and above as a standalone program and requires ~1 Gb of disk space as a standalone program. No further prerequisites are required.

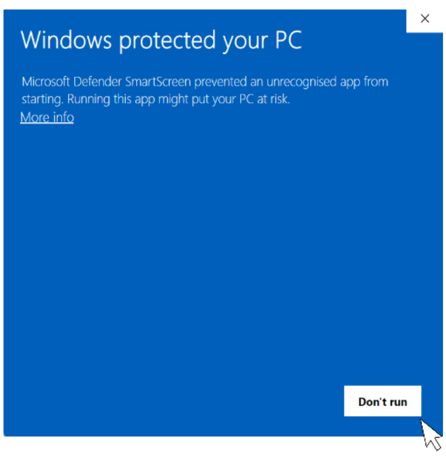
Windows:

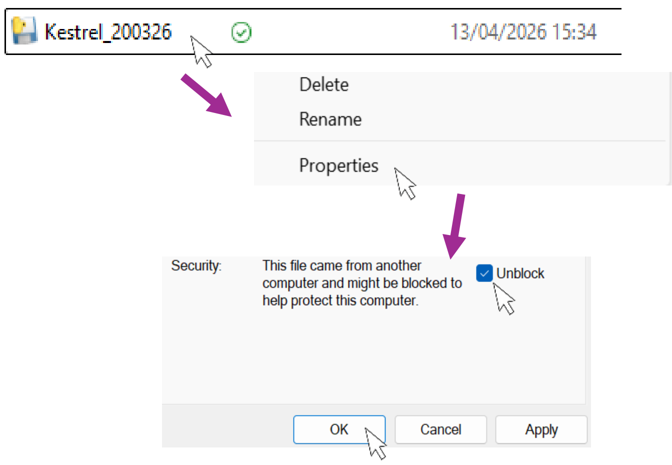
1) Download the latest version of KESTREL from our **website**. Open 'File Explorer' and navigate to your 'downloads' folder where you will see the KESTREL executable.
2) Optionally, this file can be dragged onto the desktop for ease of access.
3) Double click on the file/thumbnail to launch KESTREL.



Troubleshooting

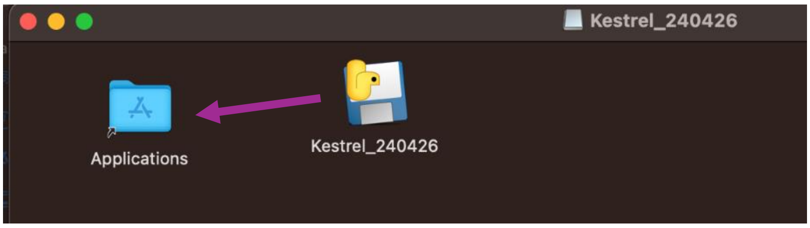
i) Depending on system security settings, the below message may be displayed.
ii) If this is the case, right click on the .exe file, select 'properties' and check the 'Unblock' box. Click 'OK' and KESTREL will run as expected.





MacOS:

1) Download the latest version of KESTREL from our **website**. Open 'Finder' and navigate to your 'downloads' folder where you will see the *Kestrel* .dmg file.
2) Double click on the .dmg file and drag the application ("*Kestrel.app*") to your 'Applications' folder (see below).
3) Double click on the thumbnail in the 'Applications' folder to launch KESTREL.

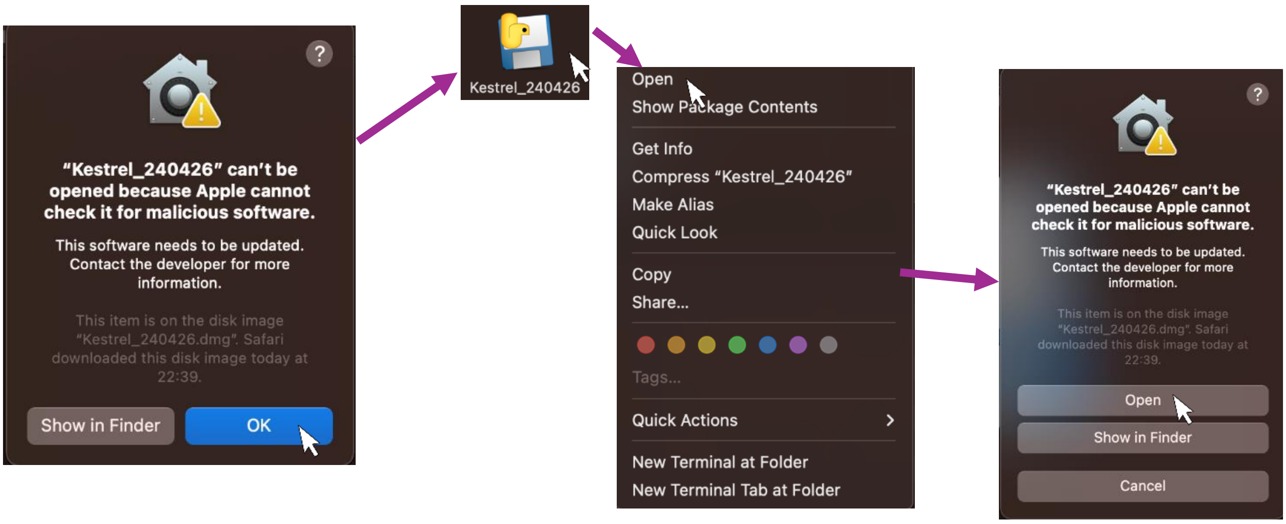


Troubleshooting

i) Depending on system security settings, you may be warned that KESTREL cannot be opened.

ii) If this is the case, click "OK" and then right/double-click on the file/thumbnail. This will open a new menu where you should select 'open'.

iii) You will then see the same warning as before but this time there will be an option to "Open". Click this and KESTREL will launch.



You are now ready to use KESTREL. The remainder of this guide contains detailed documentation on each calculation and option within KESTREL.

1.1 File and Display Options

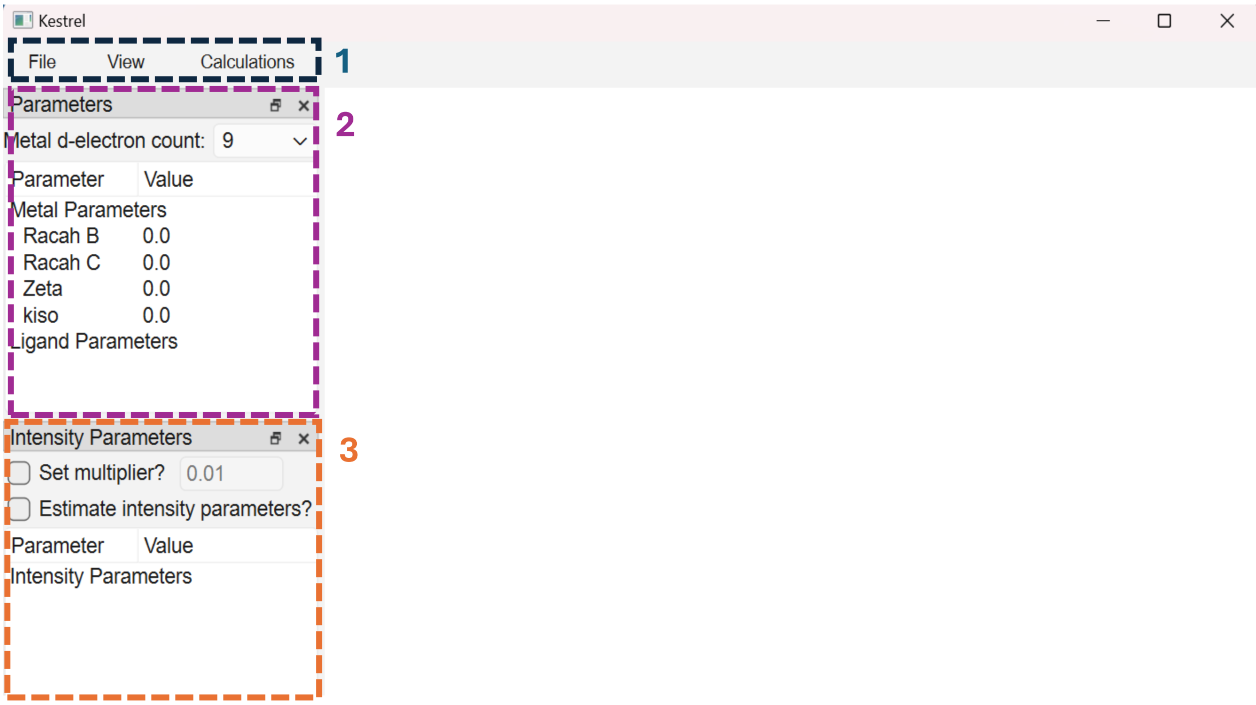
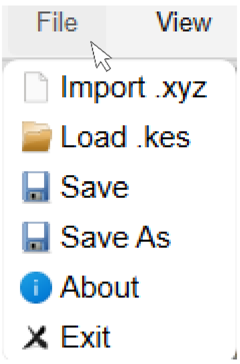


Figure 1: Appearance of KESTREL's GUI upon launching with menu options (1 - section 1.1), parameter input (2 - section 1.2) and intensity parameter input (3 - section 2.5.5) highlighted.

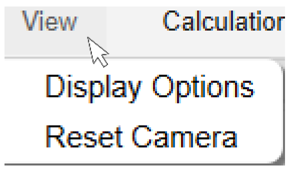
1.1.1 Menu Options

'File' reveals the following options:



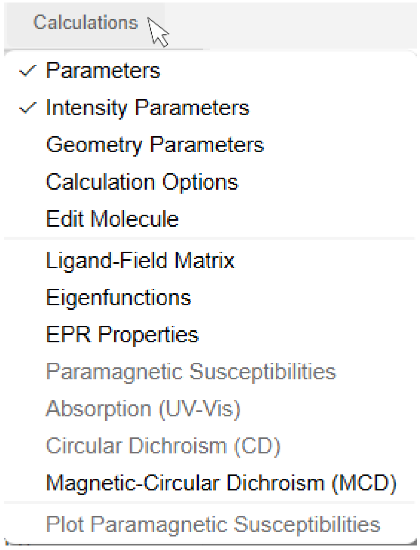
- Import .xyz file - load new molecule
- Load .kes - load KESTREL file from a previous calculation
- Save
- Save As
- About
- Exit

'View' options:



- Display options – Opens a menu (see section 1.1.2) which allows coordinate axes, ligand axes and atom labels to be shown and provides options for modifying the camera view.
- Reset camera - resets the camera to a default viewing position.

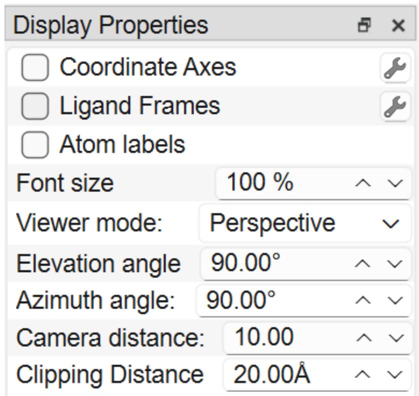
'Calculations' options:



- Parameters – enable parameters menu.
- Intensity parameters – enable intensity parameters menu
- Geometry parameters – enable geometry parameters menu
- Calculation Options – configure what calculations are run (this opens a new window).
- Edit Molecule – allows the molecule to be edited (i.e. the coordinating ligand atoms changed). Note that this will set all current parameter values to zero.
- Options to display the results of each calculation - these are greyed out unless they have been enabled in the 'Calculation Options' window with the exception of 'Ligand-Field Matrix' and 'Eigenfunctions' which are always calculated.

1.1.2 Display Options

Selecting 'Display Options' from the view menu opens the 'Display Properties' window:

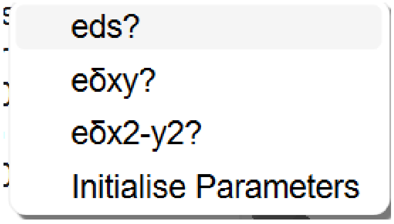


- Coordinate axes - allows the molecular xyz axes to be toggled.
- Ligand frames - allows the local ligand x/y frames to be displayed.
- Atom labels - Show labels for each atom according to its element and order in the .xyz file.
- Font size - change the font size for easier viewing.
- Viewer mode - allows perspective or orthographic views to be chosen.
- Elevation angle, Azimuth angle, Camera distance - permits manual variation of camera position.
- Clipping Distance – Toggle the distance at which atoms are no longer visible/are 'clipped'.

1.2 Parameter Options

1.2.1 Delta and e_{ds} parameters

Delta bonding can be parameterised using eδxy and eδx^2-y^2 parameters. These can be enabled for each ligand by right-clicking on the ligand name which opens a menu where the delta parameters can be selected.



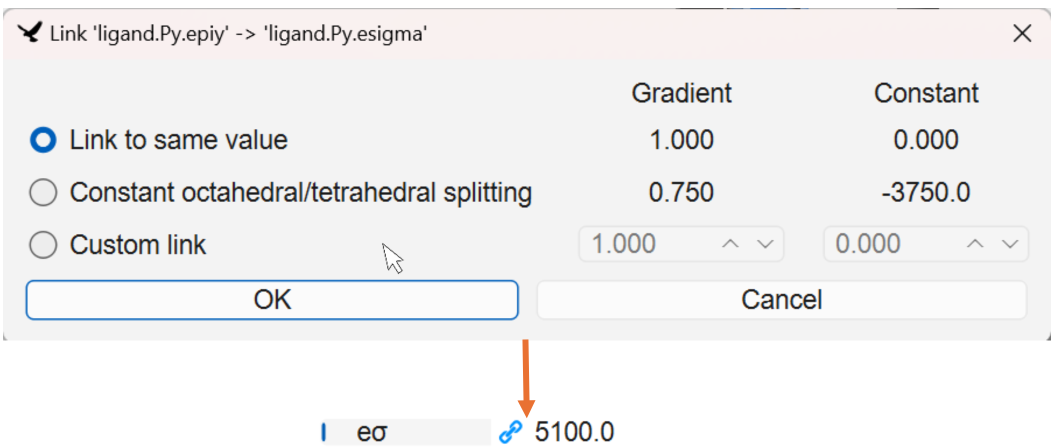
3d-4s mixing can be parameterised using the eds parameter which is enabled from the same menu.

1.2.2 Linking parameters

Parameter values can be linked by dragging a parameter value onto another parameter value which you want to link it to. This opens a menu with the following options:

- Link to same value: Sets the two parameters equal to each other.
- Constant octahedral/tetrahedral splitting: Designed for octahedral/tetrahedral complexes where multiple combinations of e_σ and e_π can lead to the same ligand-field splitting. Selecting this option links e_σ and e_π so they adjust together and keep the ligand-field splitting constant.
- Custom link: Allows a custom gradient (i.e. multiplier) and constant (i.e. offset) to be set.

It is generally useful to link parameter values when the ligands are of the same or a similar type so have similar *e* values, or for cylindrical π-donors where $e_{\pi x}$ can be set equal to $e_{\pi y}$.



1.2.3 Initialising Parameters

KESTREL has an option to estimate initial ligand parameters by right-clicking on the ligand name and choosing 'Initialise Parameters' (figure 2). The ligand type closest to the one you are parameterising should then be chosen.

Further information about each parameter type and approximate values for each are given in table 1 on page 13.

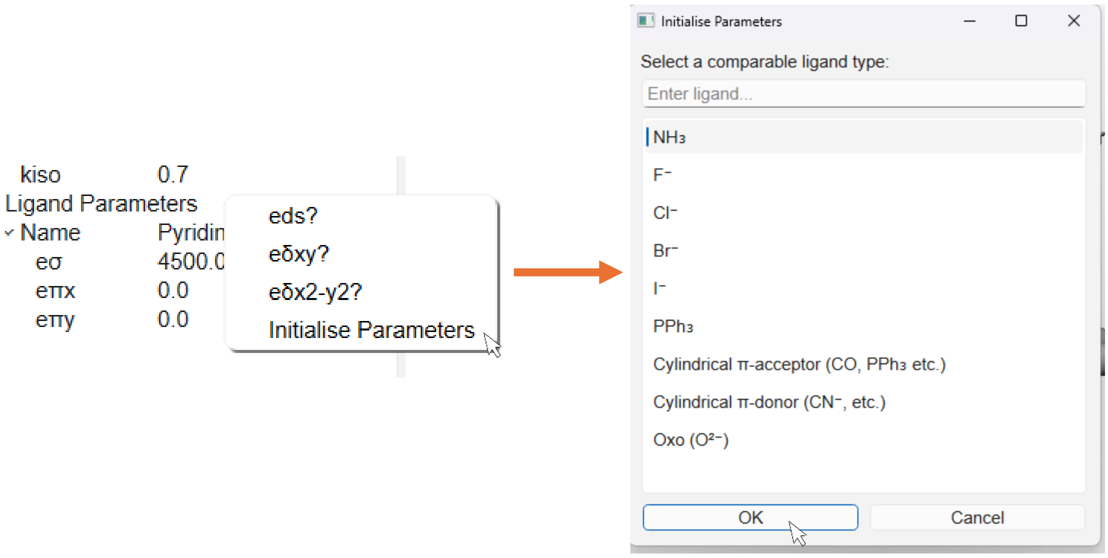
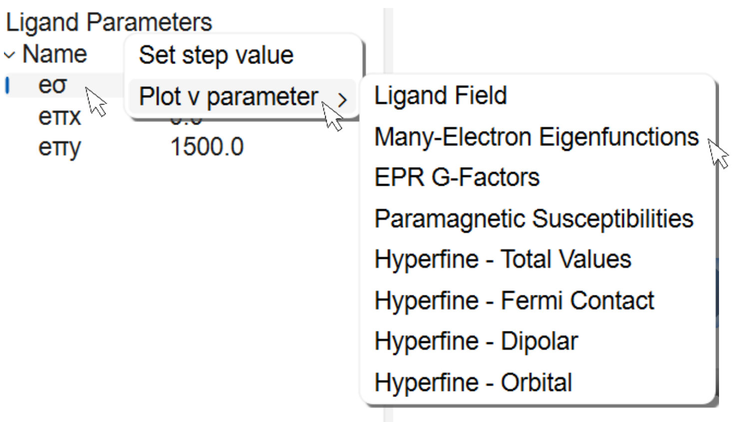


Figure 2: Initialise parameters dialog which is opened after selecting 'Initialise Parameters'. The ligand type that is most comparable to that in the complex should then be chosen to give a set of starting parameters.

1.2.4 Parameter Variation Plots - General

Right-clicking on a parameter value opens a menu where there is an option to 'Plot v parameter'. Click on this to open a further menu which allows the variation of a parameter value to be plotted against various calculation results.



As an example, we will plot the variation in eσ for $[Cu(py)_4]^{2+}$ against the energies of each electronic state ('Many-Electron Eigenfunctions'). Selecting this option opens a window where you can set a starting parameter value, a final parameter value and the number of steps. Results will then be calculated at each parameter value in this range and plotted to see how the calculation results vary with parameter input (figure 3). The graph can be panned by left-clicking, zoomed with the scroll-wheel and rescaled by right-clicking.

NOTE: A calculation must be enabled before plotting, otherwise an error message will be displayed.

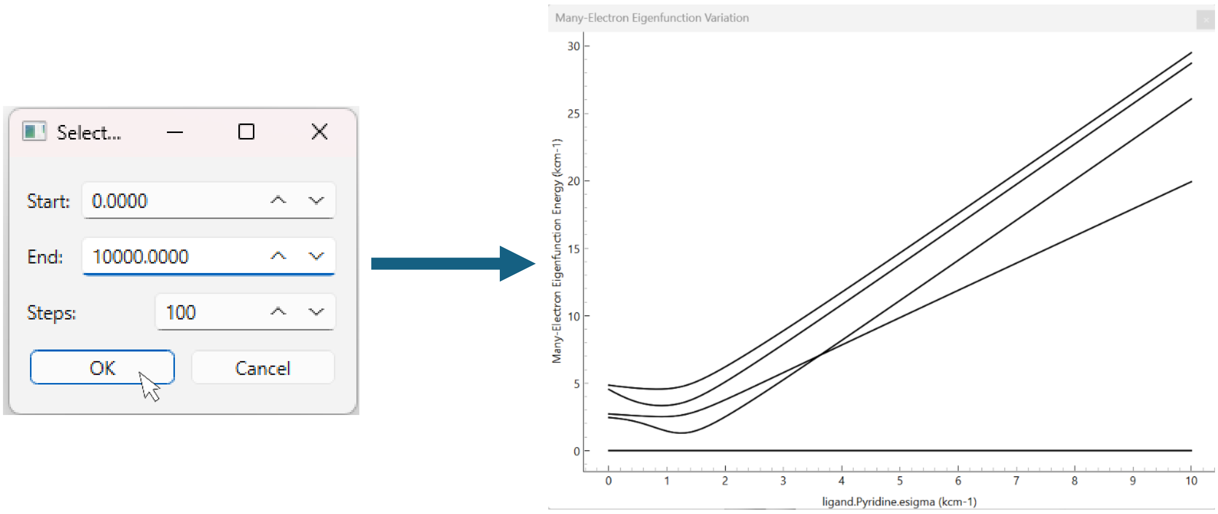


Figure 3: Variation plot of many-electron eigenfunctions for $[Cu(py)_4]^{2+}$. 5 states in total are observed as expected for a d^9 system, with their energies tending to increase in energy as the strength of pyridine σ donation increases.

Parameter variation plots can be configured by clicking the "configure graph" button (figure 4, purple box). This opens a menu where the x and y axes can be modified and the data exported as a *.csv* file or image. If changed, the view can be reset to fit the whole graph by clicking the 'A' symbol in the bottom-left corner of the window (figure 4, orange box).

1.2.5 Parameter Variation Plots – Susceptibilities

By default, the variation of the average susceptibility is plotted across a range of temperatures for each parameter value. It is also possible to plot the individual contributions (T_1, T_2 and T_3) to the susceptibility tensor by using the check boxes below the plot (figure 4, purple dashed box).

The susceptibility can also be plotted at a fixed temperature by selecting the desired temperature in the box (figure 4, orange dashed box). This then plots the susceptibility against parameter value at this temperature.

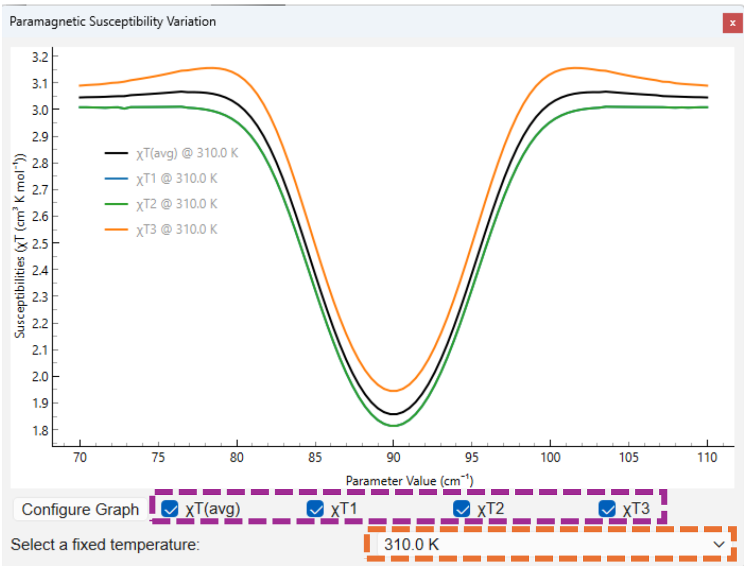


Figure 4: Susceptibility variation plot for varying the geometry (θ) (see section 1.2.7) of a d^6 spin-crossover complex.[2] Here, a fixed temperature of 310 K has been selected and the variation of both the average and individual susceptibilities has been plotted.

1.2.6 Parameter Variation Plots – Eigenfunctions

In the absence of Spin-Orbit Coupling (SOC) (i.e. if SOC/Zeta = 0 cm^{-1}), plotting the variation of many-electron eigenfunctions shows the multiplicity of each state (2S+1) in a manner akin to a Tanabe-Sugano diagram. Quintet/sextet states are coloured red, triplet/quartet states blue and singlet/doublet states black. Each spin state can be toggled on/off using the checkboxes below the graph (figure 5, black box).

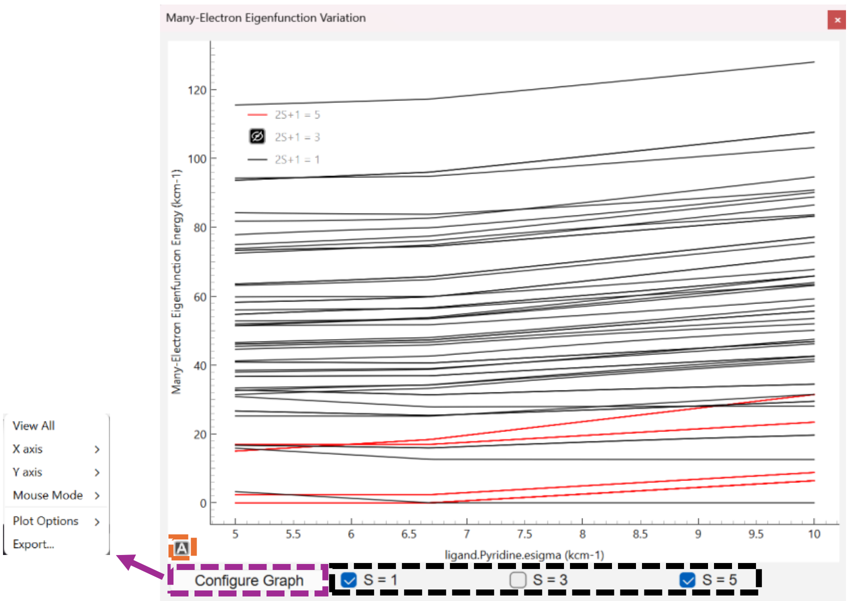


Figure 5: Variation plot of many-electron eigenfunctions for a d^6 Fe^{2+} complex[2] showing the energies of singlet and quintet states, with the triplet states hidden. Note that spin-crossover from the low-spin (2S+1=1) to high-spin (2S+1=5) states can be seen at around 6500 cm^{-1}.

1.2.7 Geometric Parameters

Geometric parameters, which describe the position of each coordinating ligand atom relative to the metal centre using Euler angles (θ, ϕ and ψ), can be modified by clicking 'Geometry parameters' in the 'calculations' menu. This opens a dock where each parameter is displayed. Each parameter value can be changed, linked and varied as for any ligand or metal parameter (see sections 2.2.2 to 2.2.5). Plotting variation graphs for each geometric quantity gives an indication of how the system changes with coordination geometry.

If a parameter value is modified, a 'dummy' atom will appear in the molecular visualiser which shows the new position of that atom (figure 6, right). Each geometric parameter can be reset to its original value by right-clicking on the parameter and selecting the 'Reset value' option (figure 5, left).

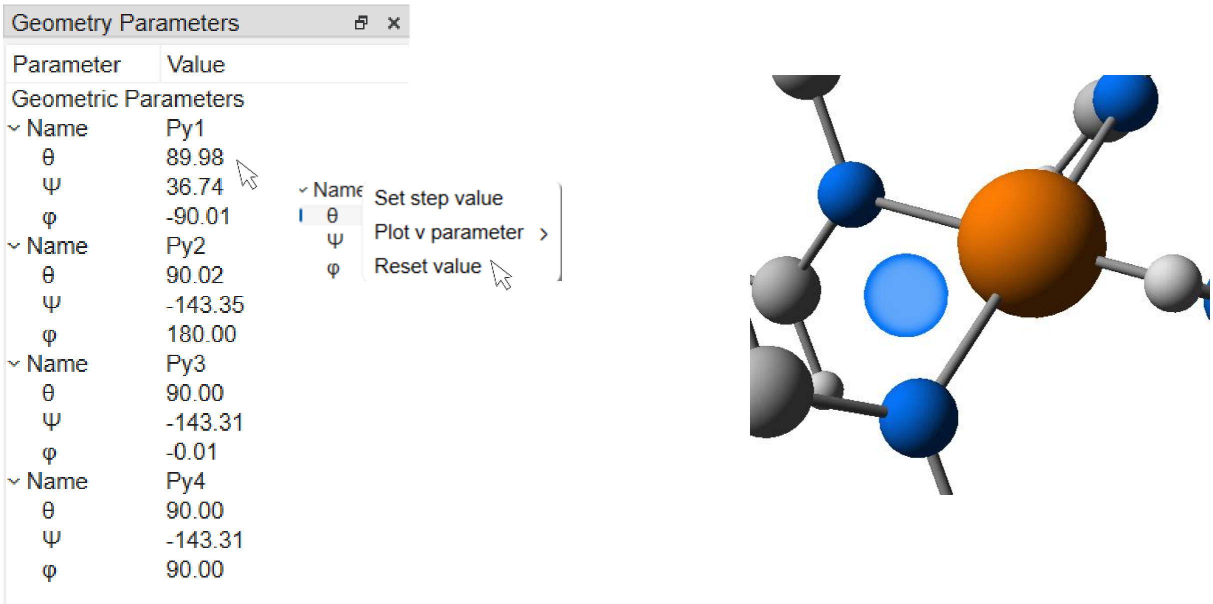


Figure 6: Left – Dock showing each geometric parameter for [Cu(py)4]2+. A parameter can be reset to its original value by selecting "Reset value" as shown. Right – Dummy atom showing the new atom position after changing θ for Py1 to 125°

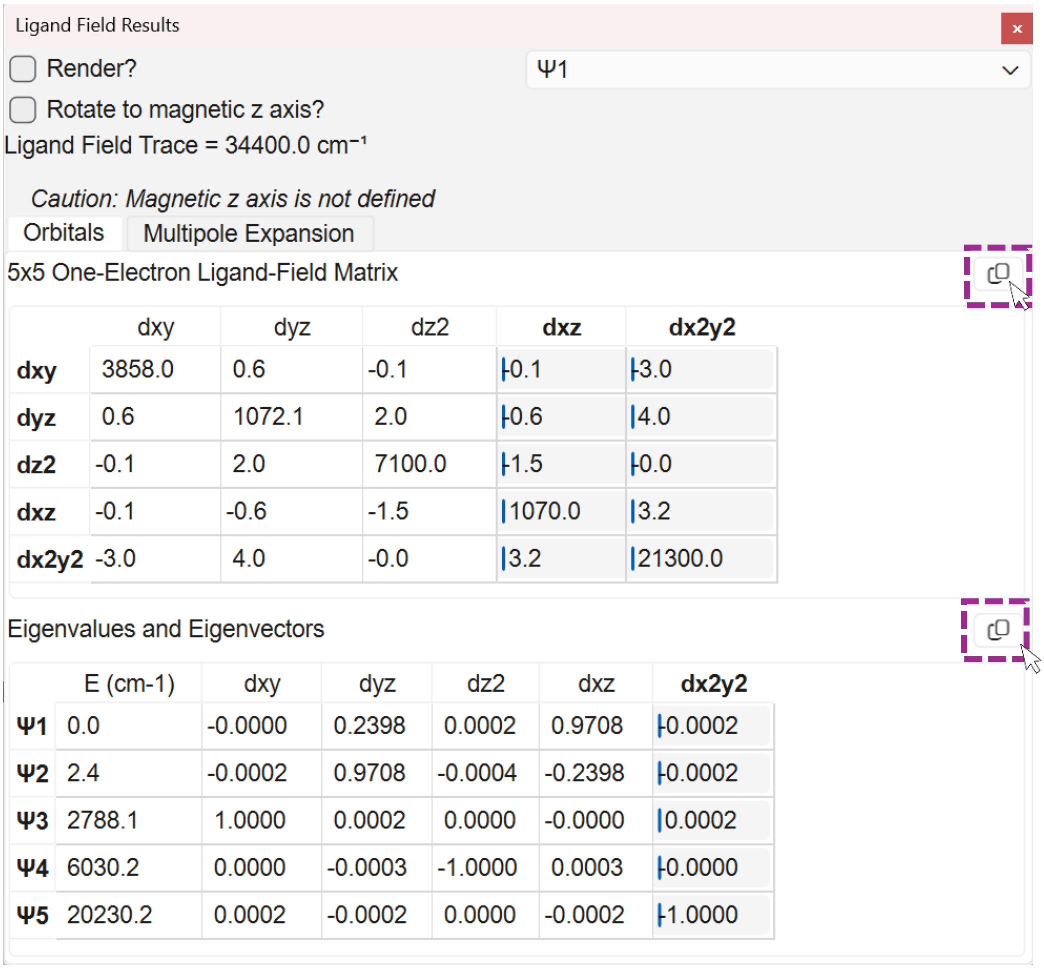
1.2.8 Table of Parameters

Parameter	Description	Indicative Range	Units
Metal parameters			
Racah B	Describes the interelectronic repulsion	0 to 1000	cm^{-1}
Racah C		approx. 4 × Racah B	cm^{-1}
Spin-orbit Coupling (Zeta)	Describes the strength of spin-orbit coupling	0 to 1000	cm^{-1}
Orbital reduction factor (k_{iso})	Describes the quenching of orbital angular momentum due to covalency effects	0 to 1	Dimensionless
Ligand Parameters			
eσ	Describes σ bonding	0 to 10000	cm^{-1}
eπx	Describes π bonding. A positive sign shows donation and a negative sign back-bonding.	-2000 to 2000	cm^{-1}
eπy			
eds	*Describes the extent of 3d-4s mixing*	*0 (optional)*	*cm^{-1}*
eδxy	*Describes delta bonding*	*0 (optional)*	*cm^{-1}*
$e\delta x^2\text{-}y^2$			
Intensity Parameters			
psigma*	Describes the polarisation of the d orbital by p orbitals. Values generally increase with decreasing symmetry.	0 to 100	Dimensionless Values are relative to each other
ppix*		0 to 50	
ppiy*			
fsigma	Describes the polarisation of the d orbitals by f orbitals. Values generally depend on the shape of the bond.	0 to 100	
fpix		0 to 50	
fpiy			
Hyperfine Parameters			
P*	Scales the strength of hyperfine interaction depending on the metal isotope.	0 to 1200	MHz or x10^{-4} cm^{-1}
Kappa*	Describes the Fermi Contact contribution	0 to 0.5	Dimensionless

Table 1: List of parameter values used in KESTREL, their definition, indicative values and units. For parameters marked with an asterix (*), KESTREL can calculate estimated values.

2.1 Ligand Field Matrix Calculation

This is enabled by default and describes the ligand field surrounding the metal centre. The output is given in two parts (figure 7). The top matrix shows the 5×5 ligand field matrix that is constructed from the inputted *e*-values. The bottom matrix gives the results of its diagonalisation. The eigenvalues (column 1) show the energies of each d-orbital and the eigenvectors (columns 2-6) the composition of each d orbital.



	dxy	dyz	dz2	dxz	dx2y2
dxy	3858.0	0.6	-0.1	-0.1	-3.0
dyz	0.6	1072.1	2.0	-0.6	4.0
dz2	-0.1	2.0	7100.0	-1.5	-0.0
dxz	-0.1	-0.6	-1.5	1070.0	3.2
dx2y2	-3.0	4.0	-0.0	3.2	21300.0

	E (cm-1)	dxy	dyz	dz2	dxz	dx2y2
Ψ1	0.0	-0.0000	0.2398	0.0002	0.9708	-0.0002
Ψ2	2.4	-0.0002	0.9708	-0.0004	-0.2398	-0.0002
Ψ3	2788.1	1.0000	0.0002	0.0000	-0.0000	0.0002
Ψ4	6030.2	0.0000	-0.0003	-1.0000	0.0003	-0.0000
Ψ5	20230.2	0.0002	-0.0002	0.0000	-0.0002	-1.0000

Figure 7: Ligand field results window showing the 5×5 ligand field matrix (top) and its eigenvalues and eigenvectors (bottom).

The tables in figure 7 can be exported to .csv files by clicking the button on the top right of each table (purple, dashed, box), further information given in the Appendix on page 31.

2.1.1 Render d orbital

KESTREL will render each d-orbital on top of the metal centre if the 'Render?' checkbox is checked. The orbital to be plotted can be selected using the drop-down menu to the right.

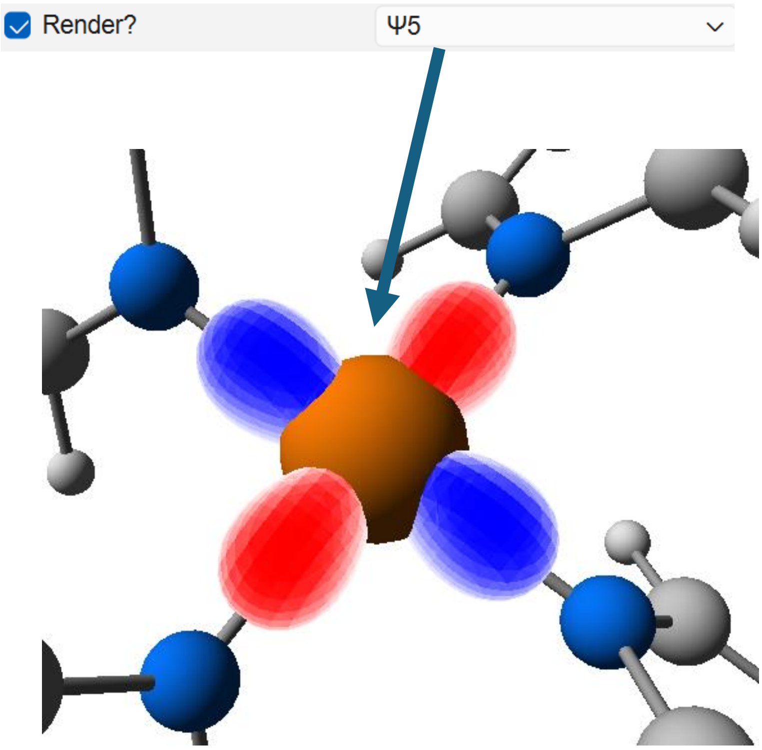
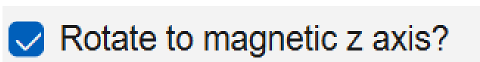


Figure 8: Render of the highest energy orbital ($\psi 5$), in this case the dx^2-y^2 orbital, on the metal centre.

2.1.2 Rotation to magnetic frame

If EPR calculations are enabled, the ligand field can be rotated into the magnetic frame. This is important for viewing the correct composition of each orbital as the initial molecular xyz frame is usually arbitrary. Checking this box will perform a unitary rotation of the ligand field matrix to align the z axis with the magnetic z axis. This ensures that the d orbital compositions are chemically meaningful.

Note: This option is only available when EPR calculations are enabled.



2.1.3 Multipole Expansion

The multipole expansion can also be viewed by selecting the ‘Multipole Expansion’ tab and rendered on the metal centre using the render option.

Orbitals | Multipole Expansion

Ckq	Real (cm-1)	Imaginary (cm-1)
2, 2	0.7	1.6
2, 1	-0.9	0.5
2, 0	-26613.0	0.0
2,-1	0.9	0.5
2,-2	0.7	-1.6
4, 4	7660.9	24238.7
4, 3	-1.5	3.9
4, 2	0.6	2.2
4, 1	-1.0	0.3
4, 0	20663.3	0.0
4,-1	1.0	0.3
4,-2	0.6	-2.2
4,-3	1.5	3.9
4,-4	7660.9	-24238.7

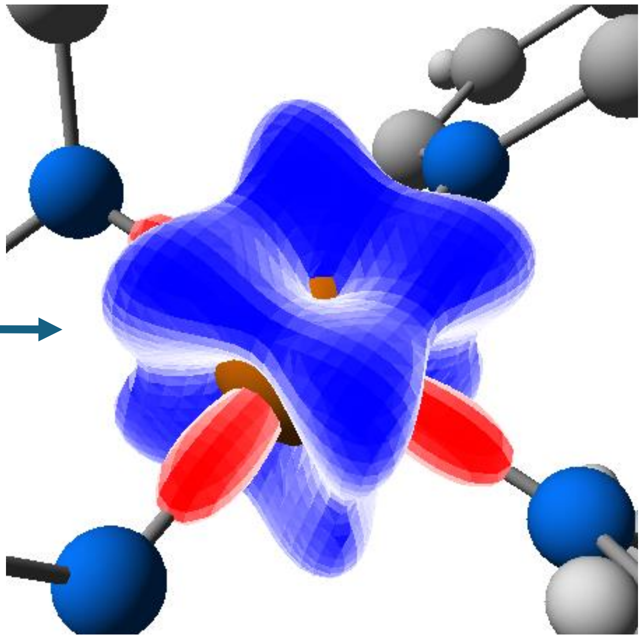
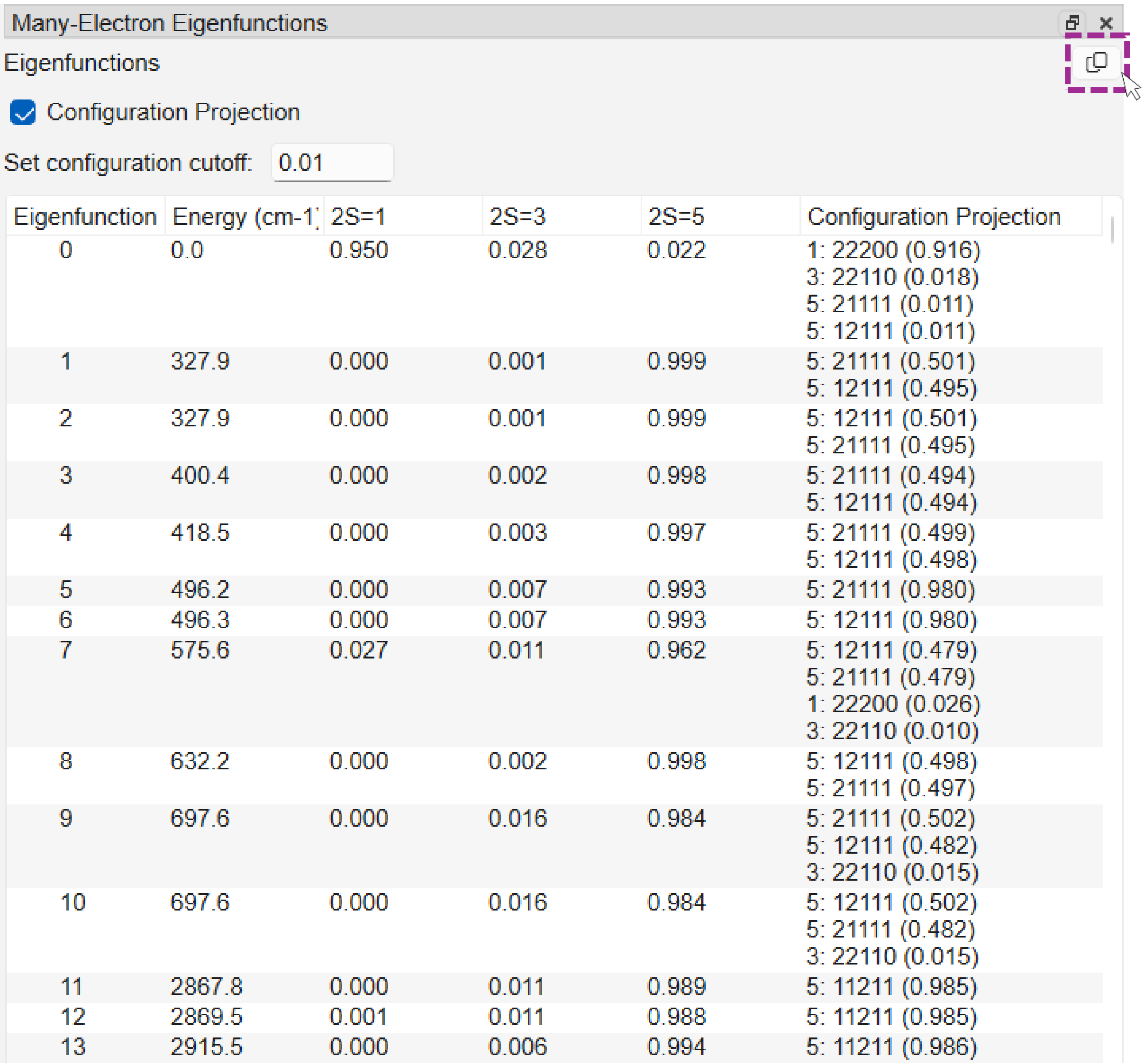


Figure 9: Render of the multipole expansion on the metal centre.

2.2 Many-Electron Eigenfunctions

This displays the energy of each electronic state and its spin projection and is enabled by default. The configuration projection can also optionally be enabled which shows the electronic configuration of the ground state and each excited state. Below is an example, with configuration projection enabled, for a d^6 Fe^{2+} complex.[2]


Many-Electron Eigenfunctions

Eigenfunctions

Configuration Projection

Set configuration cutoff: 0.01

Eigenfunction	Energy (cm-1]	2S=1	2S=3	2S=5	Configuration Projection
0	0.0	0.950	0.028	0.022	1: 22200 (0.916) 3: 22110 (0.018) 5: 21111 (0.011) 5: 12111 (0.011)
1	327.9	0.000	0.001	0.999	5: 21111 (0.501) 5: 12111 (0.495)
2	327.9	0.000	0.001	0.999	5: 12111 (0.501) 5: 21111 (0.495)
3	400.4	0.000	0.002	0.998	5: 21111 (0.494) 5: 12111 (0.494)
4	418.5	0.000	0.003	0.997	5: 21111 (0.499) 5: 12111 (0.498)
5	496.2	0.000	0.007	0.993	5: 21111 (0.980)
6	496.3	0.000	0.007	0.993	5: 12111 (0.980)
7	575.6	0.027	0.011	0.962	5: 12111 (0.479) 5: 21111 (0.479) 1: 22200 (0.026) 3: 22110 (0.010)
8	632.2	0.000	0.002	0.998	5: 12111 (0.498) 5: 21111 (0.497)
9	697.6	0.000	0.016	0.984	5: 21111 (0.502) 5: 12111 (0.482) 3: 22110 (0.015)
10	697.6	0.000	0.016	0.984	5: 12111 (0.502) 5: 21111 (0.482) 3: 22110 (0.015)
11	2867.8	0.000	0.011	0.989	5: 11211 (0.985)
12	2869.5	0.001	0.011	0.988	5: 11211 (0.985)
13	2915.5	0.000	0.006	0.994	5: 11211 (0.986)

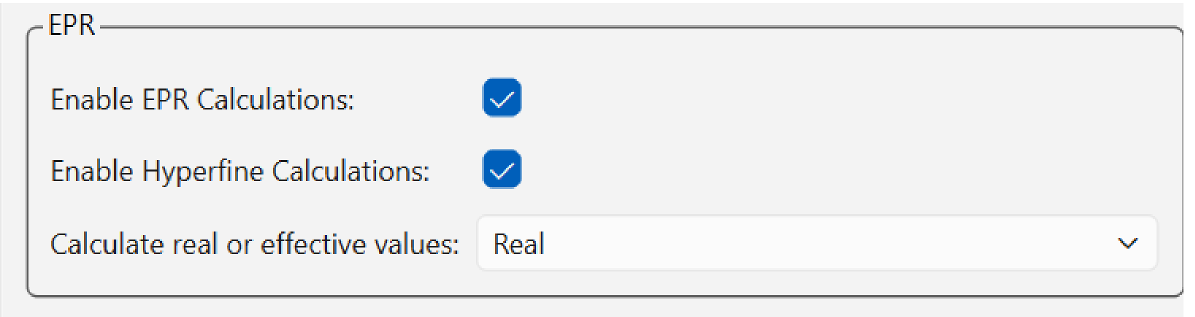
Figure 10: Many-electron eigenfunctions for a d^6 Fe^{2+} complex showing the energy of the first 9 states along with their spin multiplicity and configuration projections. From left to right, each column shows the electronic state, its energy, spin multiplicities (2S+1) (including configuration mixing), and the d orbital configuration projection.

The above results can be exported as a .csv file by clicking the icon at the top right of the table (purple, dashed, box). Further information is given in the Appendix on page 31. By default, projections contributing more than 1% to each electronic state are displayed. This can be adjusted up or down by changing the cut-off value (orange dashed box).

2.3 EPR Calculations

EPR calculations are disabled by default. To enable them, navigate to the 'Calculation Options' menu and enable EPR calculations and optionally metal-hyperfine calculations.

There is also an option to calculate 'real' or 'effective' g values which is set to real by default. This is applicable for systems with more than one unpaired electron where zero-field splitting (ZFS, see glossary) can give an effective S=1/2 system. Effective values will represent the experimental spectrum. Real values describe the actual electronic g and A values of the system and are comparable to the g and A values obtained, in conjunction with ZFS D and E parameters, from spectral simulations in software such as EasySpin.[3]



2.3.1 G tensor calculation

KESTREL will calculate g-factors for a given metal centre and outputs its orientation matrix (figure 11). There is also an option to overlay this orientation matrix to show the direction of each g value which allows the magnetic z direction to be visualised.

	g1	g2	g3
g-factor	**2.0508**	**2.0508**	**2.2619**
X	0.249	0.968	0.000
Y	0.968	-0.249	-0.000
Z	-0.000	0.000	-1.000

Figure 11: g-factors along with their orientation matrix

2.3.2 Metal-hyperfine calculation

KESTREL will calculate the metal-hyperfine coupling of an unpaired electron in a d orbital. Note that KESTREL is unable to calculate ligand-hyperfine coupling. The output gives the total A values which are observed experimentally and the orientation matrix of the A tensor (figure 12). As for the g tensor, the A tensor orientation can be overlaid (figure 13). Note that the 'real' g-factors option (see above) must be calculated for hyperfine calculations to be enabled.

KESTREL also provides information that is not readily available from experimental spectra. Namely, the sign of the hyperfine interaction, the Fermi contact (isotropic contribution), dipolar interaction (coupling between the dipole of the unpaired electron and metal nucleus) and orbital interaction representing the spin-orbit coupling contribution. This mirrors the output that is obtained from DFT calculations and allows the sign of the hyperfine coupling to be defined.

	A1 (MHz)	A2 (MHz)	A3 (MHz)
A values	**22.2**	**22.2**	**-553.8**
Fermi Contact	-311.8	-311.8	-311.8
Dipolar	276.0	276.0	-552.0
Orbital	57.9	57.9	310.0
X	-0.217	-0.976	0.000
Y	-0.976	0.217	-0.000
Z	0.000	-0.000	-1.000

Figure 12: Hyperfine values showing the principal A values and Fermi contact, dipolar and orbital contributions along with the orientation matrix. The total A values are calculated as the sum of each contribution.

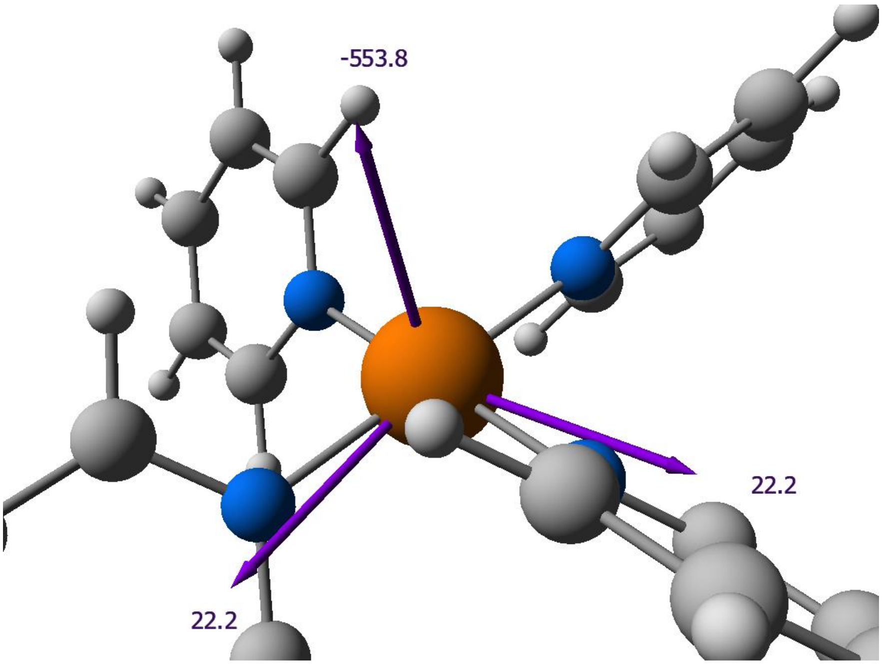


Figure 13: Overlay of the A tensor orientation

2.3.3 EPR Calculation Options

There are multiple options for EPR calculations, mainly for configuring the hyperfine calculation which are highlighted and numbered in figure 14.

1) Overlay either/both the g tensor and/or A tensor
2) Manually enter a P value which acts as a scaling factor for the hyperfine values. After checking the box, a user-defined value can be entered.
3) Manually enter a kappa value which parameterises the Fermi contact. After checking the box, a user-defined value can be entered.
4) Units for P calculation to determine if the units hyperfine is calculated with (MHz or x10^{-4} cm^{-1}). This is only relevant for cases where P is calculated automatically.
5) Specify isotope allows the metal isotope used for calculating a P value to be specified. By default, this is set to the most abundant isotope or, in the case of copper, the average of ^{63}Cu and ^{65}Cu.

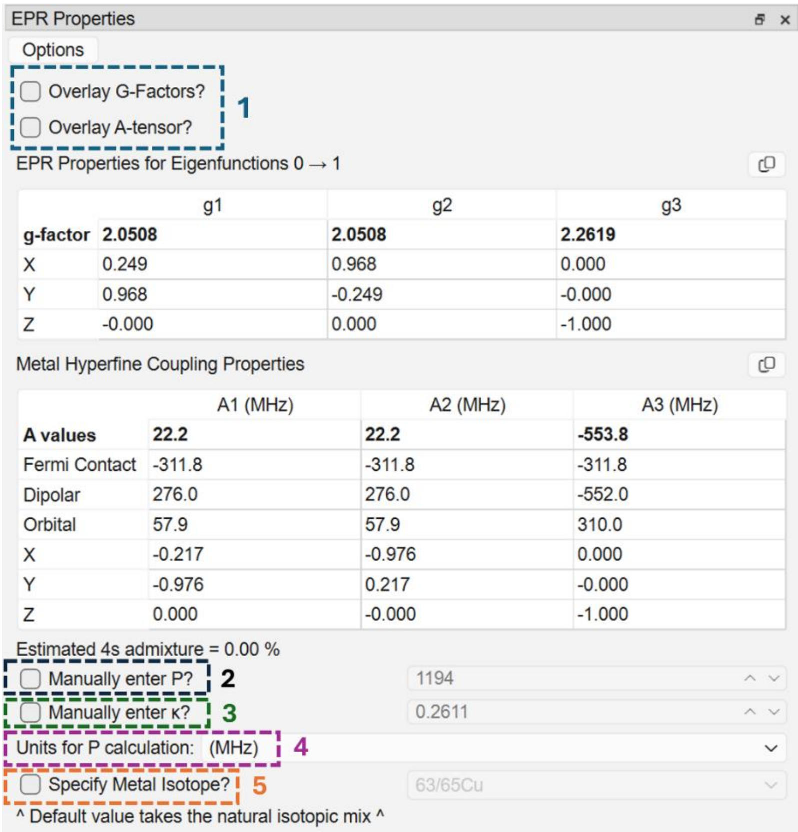


Figure 14: EPR calculation window, the calculation options detailed above are highlighted.

EPR calculation results can be exported to .csv files by clicking the icon at the top right of each table. Further information is given in the Appendix (page 31)

2.3.4 P parameter

A P value is calculated by default and acts as a scaling factor for the overall strength of hyperfine interaction.[4] It is recommended to use the calculated value unless one is not available for the metal/oxidation state being calculated or there is a physical reason for it to be reduced. The calculated value is taken from data tabulated by A.K. Koh and D. J. Miller.[5] If a tabulated P value is not available, one will need supplying manually – see FAQ (page 32).

2.3.5 Kappa parameter

Kappa is a parameter that is used to calculate the Fermi contact. Physically, it parameterises the Fermi contact interaction which arises from unpaired electron density at the nucleus, i.e. the unpaired s-electron density.[4] For a transition metal with paired core electrons, this arises from spin polarisation of the core (1s, 2s and 3s) electrons, valence-shell (4s electrons) and any direct 4s admixture into the d-orbitals.[6] KESTREL calculates kappa as the sum of these three contributions:

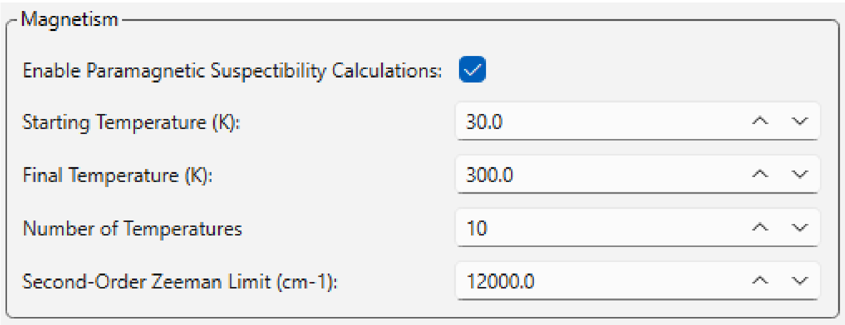
- Core-shell polarisation
- Valence-shell polarisation
- % 4s admixture

It is recommended to use the calculated value of kappa for initial fitting of experimental data. However, due to the complexity of calculating the Fermi contact, it is generally reasonable to manually vary kappa by ±10% to explore the best fit to experimental data.

2.4 Paramagnetic Susceptibilities

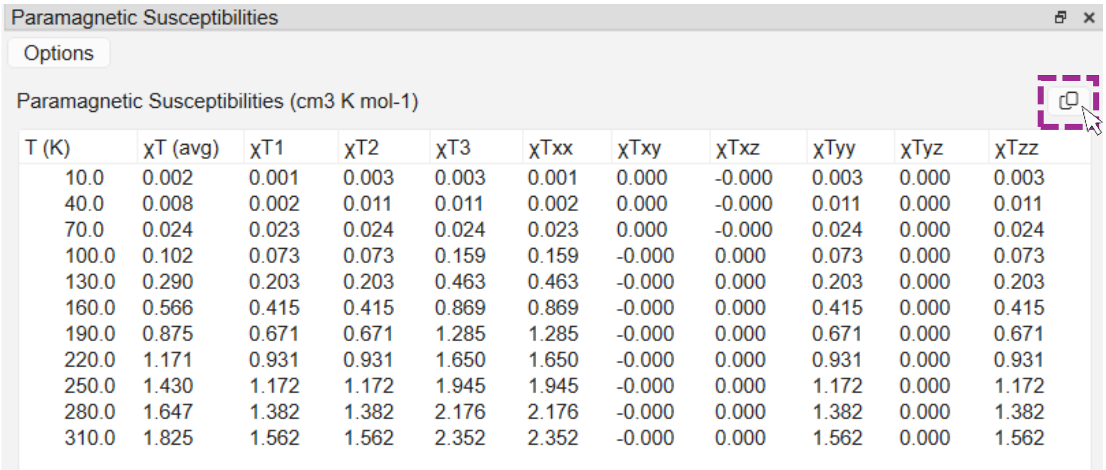
Paramagnetic Susceptibility calculations are disabled by default. To enable them, navigate to the ‘Calculation Options’ menu and enable the calculations.

There are additional options to specify the starting and final temperature susceptibilities are calculated for, the number of intermediate temperatures and the second-order Zeeman limit. The second-order Zeeman limit sets the energy cut-off for which contributions are included; the default value of 12000 cm^{-1} is appropriate for most systems.



To view the results of susceptibility calculations, select the option ‘Paramagnetic Susceptibilities’ from the ‘Calculations’ menu.

The resulting window will then display values for each direction described by the paramagnetic susceptibility tensor along with a value (2nd column) averaged over each direction.



T (K)	χT (avg)	χT1	χT2	χT3	χTxx	χTxy	χTxz	χTyy	χTyz	χTzz
10.0	0.002	0.001	0.003	0.003	0.001	0.000	-0.000	0.003	0.000	0.003
40.0	0.008	0.002	0.011	0.011	0.002	0.000	-0.000	0.011	0.000	0.011
70.0	0.024	0.023	0.024	0.024	0.023	0.000	-0.000	0.024	0.000	0.024
100.0	0.102	0.073	0.073	0.159	0.159	-0.000	0.000	0.073	0.000	0.073
130.0	0.290	0.203	0.203	0.463	0.463	-0.000	0.000	0.203	0.000	0.203
160.0	0.566	0.415	0.415	0.869	0.869	-0.000	0.000	0.415	0.000	0.415
190.0	0.875	0.671	0.671	1.285	1.285	-0.000	0.000	0.671	0.000	0.671
220.0	1.171	0.931	0.931	1.650	1.650	-0.000	0.000	0.931	0.000	0.931
250.0	1.430	1.172	1.172	1.945	1.945	-0.000	0.000	1.172	0.000	1.172
280.0	1.647	1.382	1.382	2.176	2.176	-0.000	0.000	1.382	0.000	1.382
310.0	1.825	1.562	1.562	2.352	2.352	-0.000	0.000	1.562	0.000	1.562

Figure 15: Results of paramagnetic susceptibility calculation for a d^6 Fe^{II} complex,[2] its electronic eigenfunctions are given in section 2.4.1.

The data in the above table (figure 15) can be exported as a .csv file by clicking the icon at the top right of the table (purple, dashed, box). Further information is given in the Appendix (page 31).

2.4.1 Plotting susceptibility curves

Magnetic susceptibilities can be plotted against the temperature by navigating to the 'Calculations' menu and selecting 'Plot Paramagnetic Susceptibilities'. The below figure shows the susceptibility curve for a d^6 spin-crossover complex,[2] which exhibits crossover from the low-spin to high-spin state at the plot's midpoint (~250K).

The graph can be panned by left-clicking, zoomed with the scroll-wheel and rescaled by right-clicking. The contributions to the susceptibility tensor can also be toggled using the check boxes below the graph, by default only the average susceptibility is plotted.

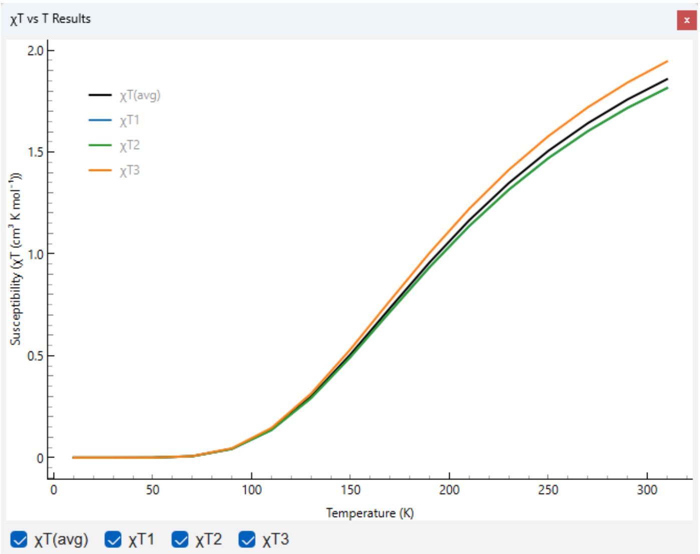
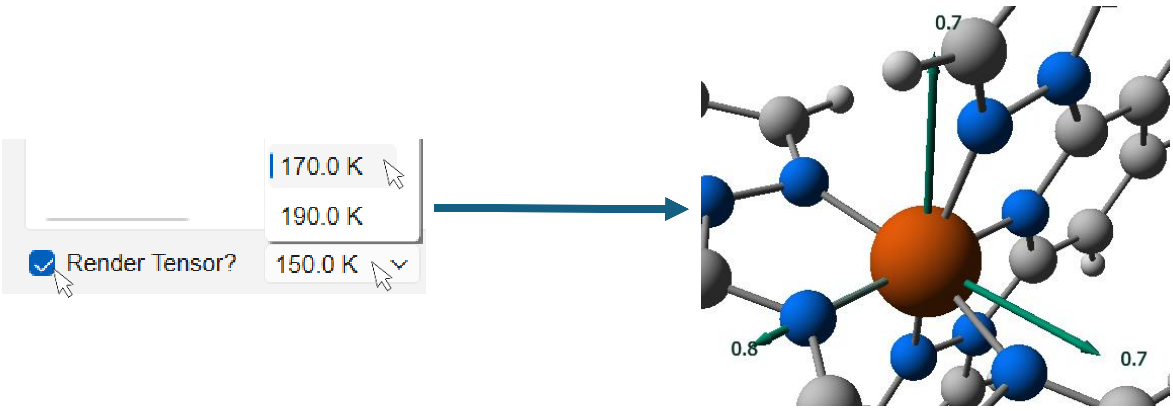


Figure 16: Paramagnetic susceptibility curve plotted from the data in figure 14. Here, the average susceptibility along with each contribution have been plotted.

2.4.2 Plotting the Susceptibility Tensor

The orientation of the susceptibility tensor (χ) can be plotted to visualise its orientation with respect to the molecular geometry. This is achieved by checking the box titled 'Render Tensor?'. This then enables a box where the temperature at which the tensor is plotted can be specified. The susceptibility tensor is then displayed on screen.

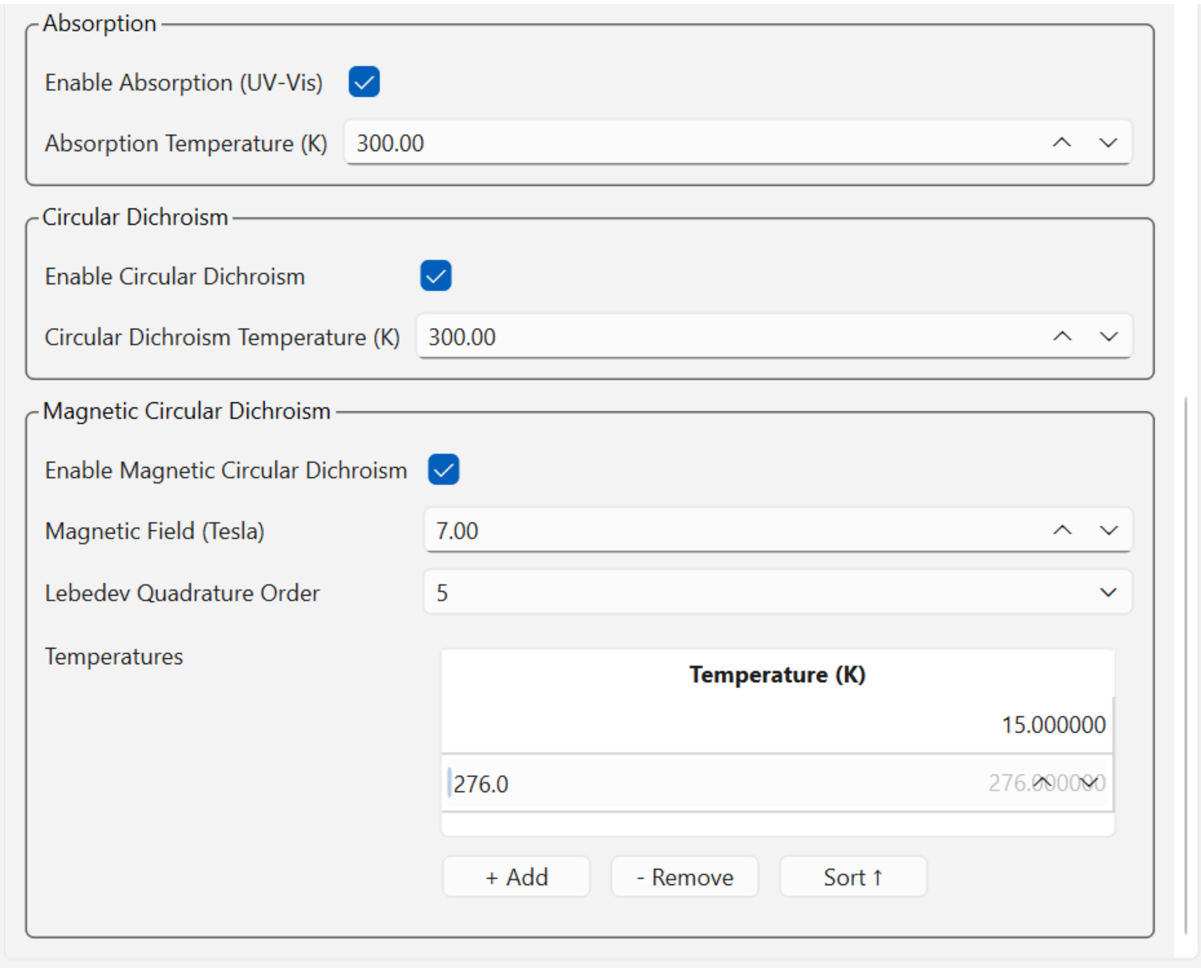


2.5 Simulating electronic spectra (UV/CD/MCD)

The simulation of electronic spectra is disabled by default. To enable these calculations, navigate to the 'Calculations' menu and check the options to enable Absorption, Circular Dichroism (CD) and Magnetic Circular Dichroism (MCD) spectra, including A, B and C terms as required. For Absorption and CD spectra, the temperature can also be specified. For MCD options, see section 2.5.1. KESTREL will calculate the d-d transition energies and their intensities, however charge-transfer bands are not calculated and are therefore absent from the simulated spectra.

For spectra to be calculated, a set of intensity parameters need to be entered in the Intensity Parameters dock (if this is not open navigate to 'Calculations' then select 'Intensity Parameters').

Note: KESTREL has a utility that will estimate the intensity parameters, see section 2.5.5 for more details on this.



2.5.1 MCD Options

The simulation of MCD spectra takes additional options compared absorption and CD spectra:

- **Magnetic field:** Allows the magnetic field strength to be specified. This can be adjusted to match the experimental field strength.

- **Lebedev Quadrature Order:** The calculation of a MCD spectrum requires the spectrum to be calculated at various points on a sphere and averaged. This process is optimised by using a Lebedev quadrature, the order of which can be specified here.
  - The order can be varied from 3 to 31, corresponding to a range of 6 to 350 points on a sphere. By default, order of 3 (6 points) is used.

  The higher the order, the more accurate the calculated spectrum as more orientations are averaged, but this is at the cost of computational performance. For general use, an order of 3 is sufficient with higher orders (i.e. 13) recommended for publication quality spectra.

- **Temperatures:** MCD spectra are highly temperature dependent and are often measured at multiple temperatures experimentally. Here, multiple temperatures can be added with the MCD spectrum calculated at each. Figure 17 shows an example of a MCD spectrum calculated at multiple temperatures.

2.5.2 Displaying Spectra

To display a simulated spectrum, navigate to the ‘Calculations’ menu and select the spectrum to be viewed. This opens a window where the spectrum is plotted. Figure 17 shows the absorption and MCD spectrum for an Fe^{IV}(O) d^4 species.[7] The MCD spectrum has been calculated using a Lebedev quadrature of order 3 (6 points), a magnetic field of 7 T and at temperatures of 2, 10, 20 and 80 K. The absorption spectrum has been calculated at 300 K.

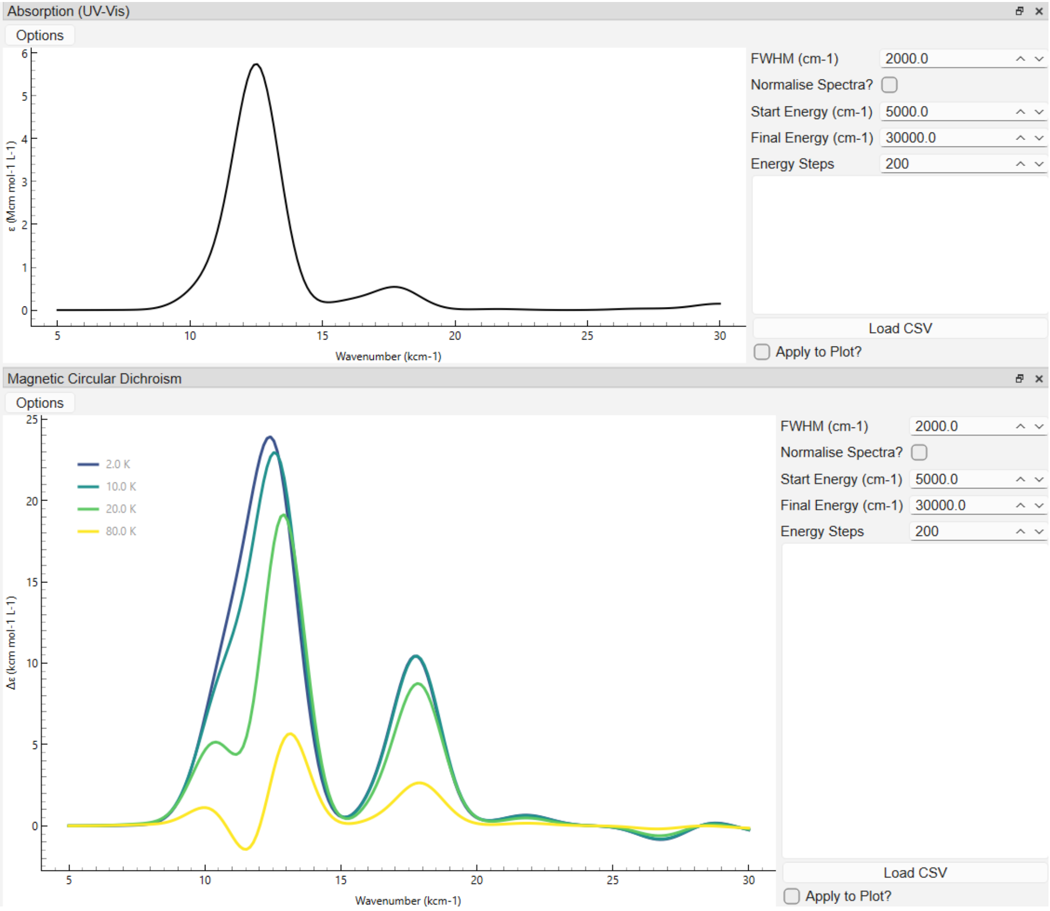
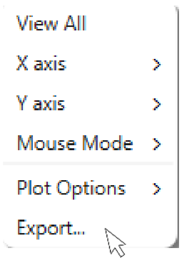


Figure 17: Windows showing the results of MCD and absorption (d-d) spectrum simulation in KESTREL for a Fe^{IV}(oxo) complex.[7]

2.5.3 Exporting Spectra

Spectra can be exported by right-clicking anywhere on the spectrum. This should then display a menu where the graph can be exported to a filetype of your choice. Selecting the ‘Export…’ option will open another menu where the filetype and save location can be specified.



2.5.4 Spectrum display options

To the right of each spectrum are various display options:

- Full Width at Half Maximum (FWHM): Set the FWHM, i.e. width, of a peak.
- Normalise spectra: Allows the spectrum to be normalised. This is useful when overlaying an experimental spectrum (see section 2.5.4).
- Start and Final Energy: Specify the range, in wavenumbers, that is displayed.
- Energy steps: Number of points that are plotted.

Each graph can also be panned by left-clicking, zoomed with the scroll-wheel and rescaled by right-clicking.

2.5.5 Loading experimental spectra

KESTREL has an option to overlay an experimental or previously calculated spectrum from a .csv file. This permits comparison to and fitting of experimental spectra.

To load a .csv file, click the 'Load CSV' button at the bottom of the right panel, which will allow you to navigate to and open a .csv file. This will import the data which can then be applied to the displayed spectrum by checking 'Apply to Plot?'. This overlays the spectrum as a dashed line (red for absorption and CD, orange for MCD) as in figure 18 so the spectra can be compared. The format of .csv files accepted by KESTREL is detailed in Appendix 1 (page 31).

Note: As KESTREL does not calculate charge-transfer transitions, it is only possible to fit to d-d transitions in experimental spectra.

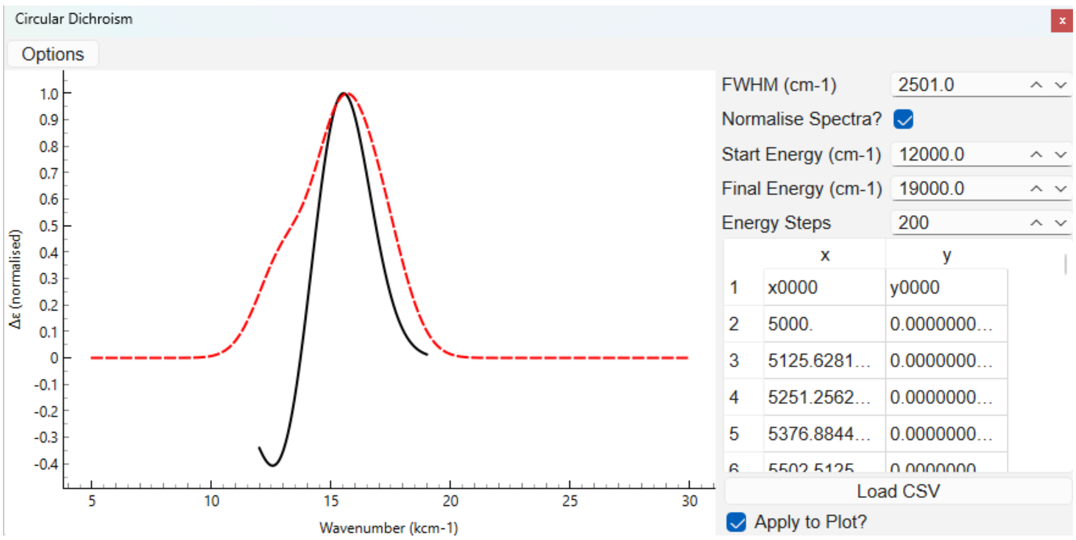


Figure 18: Overlay of simulated CD data (black) and the experimental CD spectrum (red). Parameter values can then be fine-tuned to obtain the best fit to the experimental spectrum.

2.5.6 Predicting Intensity Parameters

KESTREL can estimate the pσ, pπy and pπx intensity parameters required to simulate electronic spectra. Parameters can be estimated for all ligands and updated in real-time as the *e* parameters are varied by checking the box 'Estimate intensity parameters' in the Intensity Parameters window (see figure 19). Alternatively, p parameter values can be initialised for each ligand individually by right-clicking on the ligand name and choosing the option 'Estimate p Parameters'. Note that values initialised in this way will not update as the *e* parameters are changed.

Since a larger p value indicates a significantly different bonding environment opposite to a ligand, whilst an identical bonding environment (i.e. a perfectly octahedral $[Mn(H_2O)_6]^{2+}$ complex) will have a value near zero, KESTREL estimates a value by determining the difference in the effective ligand field at each ligand and the position directly opposite. Fσ, fπx and fπy parameters currently need entering manually and describe the 'width' of a bond. We would recommend that these values are fixed initially (i.e. at 0) and used when fine-tuning a simulated spectrum. This calculation is only intended to give approximate simulations; it is entirely reasonable to vary each parameter and add f parameters to obtain closer fits to experimental spectra.

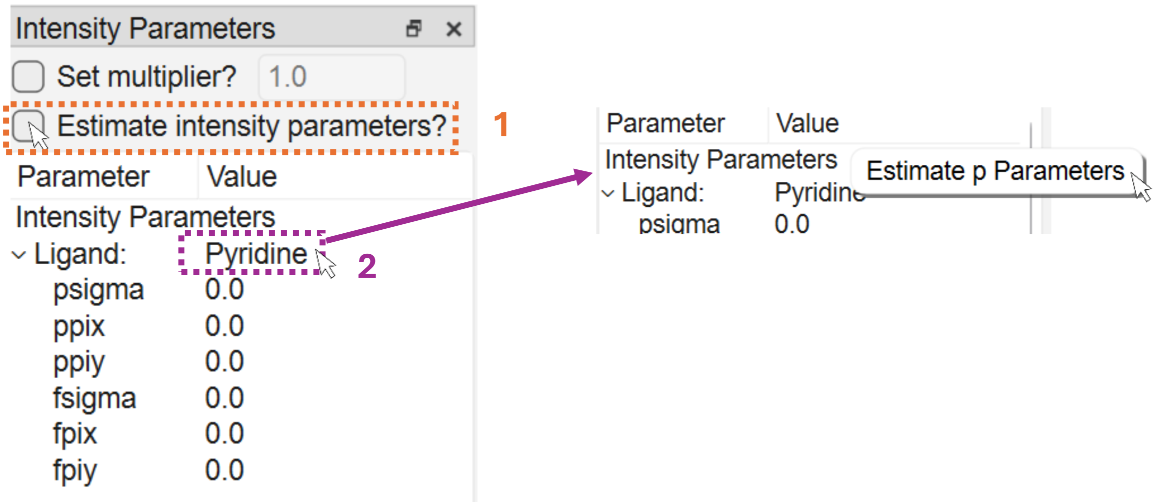


Figure 19: Intensity parameters can be estimated for all ligands and updated in real-time by checking the 'Estimate intensity parameters?' box (1) or estimated for each ligand individually (2).

Figure 20 shows MCD spectra at 10K (left) and 40K (right) that have been simulated using KESTREL's estimated intensity parameters for an Fe^{IV}(oxo) complex.[7] The dashed orange overlays show MCD spectra simulated by adjusting the intensity parameters to fit experimental data and the purple lines spectra predicted with estimated p values. Clearly, there is good agreement between the parameters estimated by this utility and parameter values that have been fitted manually.

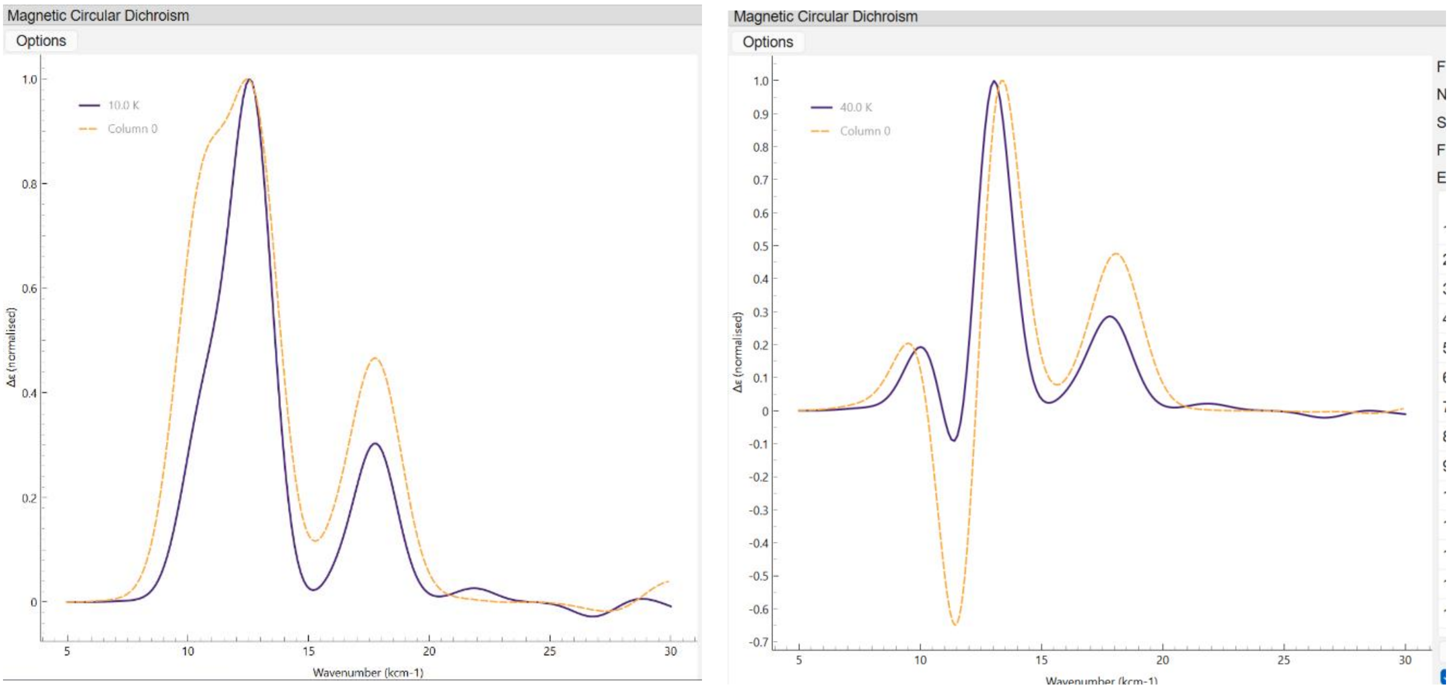
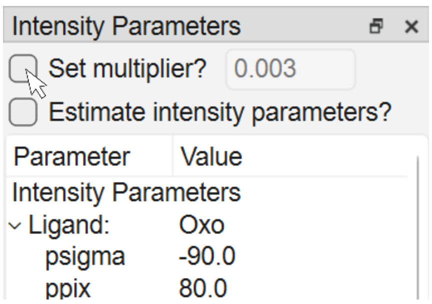


Figure 20: MCD spectra at 10K (left) and 40K (right) with intensity parameters determined via KESTREL's estimation in purple (solid line) and via fitting to the experimental MCD spectrum in orange (dashed line).

2.5.7 Intensity Parameters Multiplier Value

The intensity parameters entered in KESTREL are generally treated as relative, rather than absolute values. This approach gives the correct shape of a spectrum, however, to calculate accurate absolute molar extinction coefficients (ε), values should be set relative to a value of ~1. To aid in this, KESTREL has an option to apply a multiplier, which scales each intensity parameter value. If desired, this multiplier can be adjusted to obtain 'correct' absolute ε values. To set a multiplier value, check the box titled 'Set multiplier?' and enter the desired value. However, it should be stressed that setting a multiplier is not necessary for replicating the 'shape' of each spectrum.

When predicting p parameter values (see last section, 2.5.5), the multiplier is calculated automatically so that the value of the value of the largest p parameter is 1.



Appendix 1 – Supported filetypes

.xyz file

This filetype is used to describe a molecular structure that is to be imported into KESTREL. It specifies each element and defines its position using xyz coordinates.

An .xyz file can be generated with common molecular visualisation software (i.e. Avogadro, Mercury and Pymol) from crystal structures by saving with the .xyz file extension or by hand for idealised structures such as octahedral or square planar complexes. Before loading a .xyz file it is important to ensure it matches the format below.

```
CuCl4
5
Cu    0.00    0.00    0.00
Cl    1.80    0.00    0.00
Cl   -1.80    0.00    0.00
Cl    0.00    1.80    0.00
Cl    0.00   -1.80    0.00
```

The first two lines are optional for KESTREL but are often required by other visualisation software. The first line describes the structure and the second line the number of atoms. Each atom is then listed in turn with its element symbol followed by x/y/z position in Å separated by blank spaces. The coordinates above describe a square planar $CuCl_4$ complex.

.kes file

This filetype stores a previous KESTREL calculation. When a calculation is saved, a .kes file is generated which stores the current parameter values and calculation results. This allows calculations to be shared with others and returned to for future analysis.

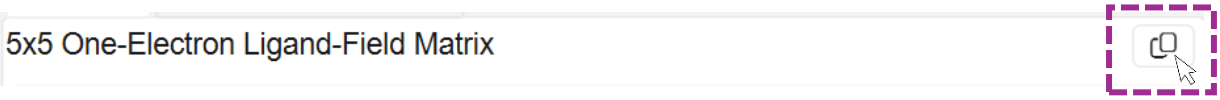
When loading a previous session, a .kes file will be required. If no previous calculations have been run, a .xyz file will need importing instead – see previous section.

Example .xyz and .kes files can be found here for familiarisation with the program and to compare results to.

.csv file

Exporting

KESTREL will export the results of calculations as .csv files which allows results to be analysed or plotted in workbooks. A .csv file can be generated by clicking the below button (purple, dashed, box) which can be found adjacent to each calculation result.



In this example, the 5x5 Ligand Field Matrix will be exported to a .csv file which will be ordered as below and represents the ligand field matrix seen in KESTREL.

<Di|V|Dj>	dxy	dyz	dz2	dxz	dx2y2
dxy	3858	0.6	-0.1	-0.1	-3.1
dyz	0.6	1072.1	2.1	-0.6	4.2
dz2	-0.1	2.1	7400	-1.6	0
dxz	-0.1	-0.6	-1.6	1070	3.4
dx2y2	-3.1	4.2	0	3.4	22200

Importing

Comparison MCD, UV and CD spectra (see section 2.5) can be loaded as .csv files and compared to the spectra simulated by KESTREL.

For this the below file format is required with this example showing the UV spectrum for LsAA9, a copper-containing metalloprotein.[8]

X0000	y0000
10000	0.00313
10075.38	0.003859
10150.75	0.004733
10226.13	0.005777
10301.51	0.007015
10376.88	0.008477
10452.26	0.010192
10527.64	0.012193
10603.02	0.014514

Note the first column contains the x value (wavenumber, cm^{-1}) and the second column the y value (absorption, dm^3 mol^{-1} cm^{-1}). It is important that imported .csv files are in this format.

Frequently Asked Questions

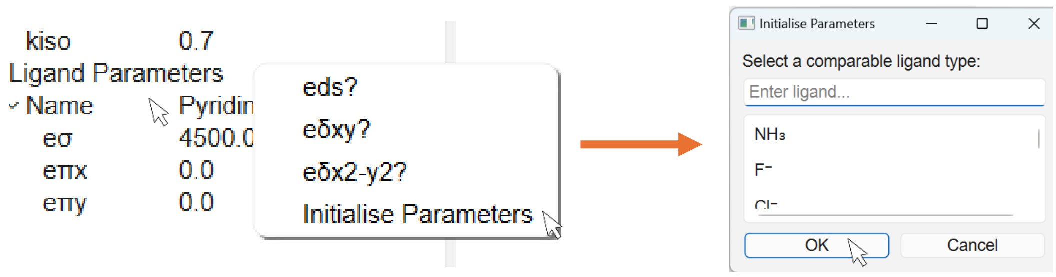
What if I don't have an .xyz file?

- If you have access to a crystal or protein structure file (.cif, .mol2 etc.) then import this into a molecular visualisation program such as mercury or Avogadro. The structure can then be saved as a XYZ file by selecting this filetype.
- Alternatively, for simple octahedral or square planar complexes, xyz files can be generated by hand following the format in Appendix 1.
- Finally, if no crystal structure is available, DFT optimisation will generate an xyz file that can be directly imported into KESTREL.

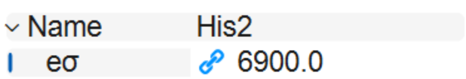
What is the best way to parameterise a complex?

We suggest the following workflow once a ligand has been imported, although this can be adapted depending on personal preference and experience:

1) Initialise parameter values for each ligand (section 2.1.5) by choosing the most comparable. This gives a set of initial parameter values.



2) Link similar ligand types together (see section 1.2.2) to reduce the number of parameters being varied. Later, they can be unlinked for fine-tuning.



3) Set Racah B and SOC/Zeta to free-ion values and Racah C to be 4*Racah B.
4) Adjust eσ values to get an approximate fit to experimental data. Do not worry if this fit does not match the 'exact' experimental values.
5) Repeat step 4 and start adjusting eπ values as well IF the ligand forms π bonds, to more closely match experimental data. Iterate this process to get a good fit to experimental data.

Finally, check that the parameter values make chemical sense. Since they represent the strength and type of bonding their strength can be compared against the position of the ligand in the spectrochemical series and expected σ-donor and π-donor/π-acceptor properties.

Are there any sources of reasonable *e* parameter values?

Yes, see our website for a table of parameter values taken from previously published ligand field calculations. This table is sortable and can be searched by metal and ligand. The values here provide a starting point for ligand field calculations.

https://chem-kestrel.github.io/#parameters

Search metal or ligand type... All metals 46 entries

Cr	+3	NCS^-	6,240	380	380	Hoggard (2004) ↗
Cr	+3	CN^-	8,310	-290	-290	Hoggard (2004) ↗
Cr	+3	Pyridine	5,800	0	-580	Hoggard (2004) ↗
Cu	+2	Histidine	6,200	0	1,450	Nunn (2023) ↗
Cu	+2	NH_2	5,500	0	0	Nunn (2023) ↗
Cu	+2	F^-	5,300	2,350	2,350	Buchhorn et al. (2022) ↗

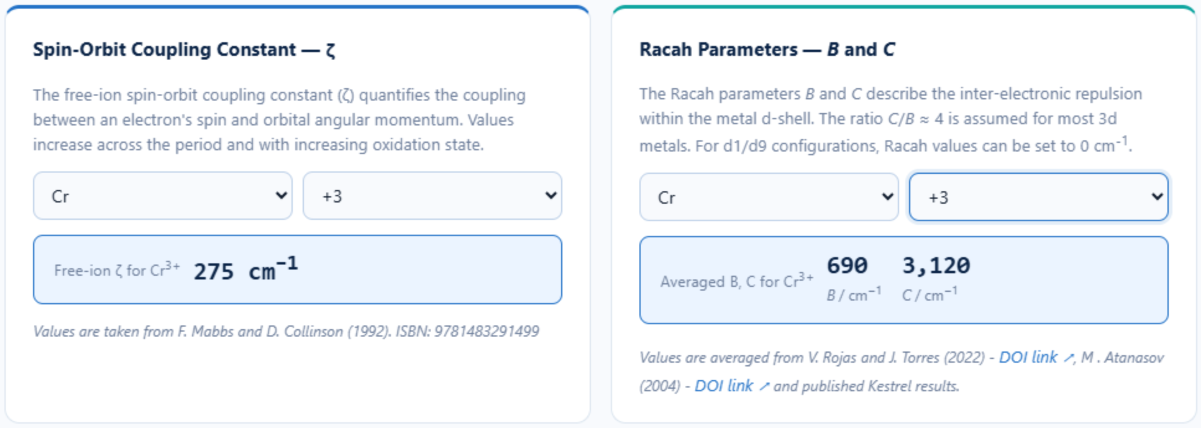
Show all entries

You can alternatively initialise parameter values within KESTREL's GUI as detailed in section 1.2.3. If there are no parameters tabulated for a ligand, then *e* values for a chemically similar ligand should be used (i.e. pyrazolyl for pyridine) and adjusted according to the actual ligand's position in the spectrochemical series.

Are there any sources of metal parameter values?

Yes, our website contains a tool where the metal and oxidation state can be specified. Free-ion spin orbit coupling (SOC) values are then given along with the average Racah B/C value for complexes with the same metal centre and oxidation state. If a parameter value is not available for a given metal and oxidation state combination, we recommend choosing the next closest as an initial estimate. The exact values obtained from this tool can then be adjusted during the fitting process.

https://chem-kestrel.github.io/#parameters



As a general guide, we recommend using free-ion values for SOC initially. Racah C can also be assumed as 4*B for initial fitting. For d1/d9 systems, the interelectronic repulsion can be ignored as will not affect the calculation results.

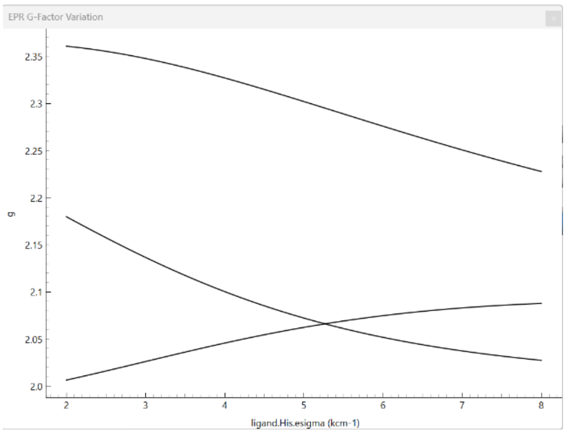
Are there any sources of intensity parameter values?

KESTREL contains a function to estimate these parameter values that will provide a good first approximation. Their precise values can then be fine-tuned to fit experimental spectra. Generally, a p value will increase as the coordination environment opposite becomes increasingly different, i.e. a perfectly octahedral species will have a p value of ~0. F values describe the width of the bond.

What if it is difficult to obtain a good fit to experimental data?

There are a couple of approaches to try in this scenario to help determine a set of physically reasonable parameters that fit the experimental data:

1) Plot variation graphs for each parameter against the results of a ligand field calculation. Details of how to do this are in section 1.2.4. This gives a visual representation of how experimental results depend on the parameter value and can help guide which values to use. To the right is an example variation plot for the EPR g factors for LsAA9, a Cu^{2+} metalloprotein.



2) Use separate parameter values for ligands of the same type. For initial fitting, it is good to use common parameters for each ligand type, for instance, if there are two Cl^- ligands in a complex the same values should be used. However, there are multiple reasons why their bonding may differ so using different parameter values should be investigated if a fit is proving elusive.

3) Check the molecular structure carefully. KESTREL is sensitive to the precise angular arrangement of the ligands, if an idealised geometry is used then this may lead to difficult in fitting to experimental data.

4) Finally, there may be elements of delta bonding or 3d-4s mixing depending on the system being analysed. In this case, these parameters can be enabled as detailed in section 1.2.1 and 1.2.2.

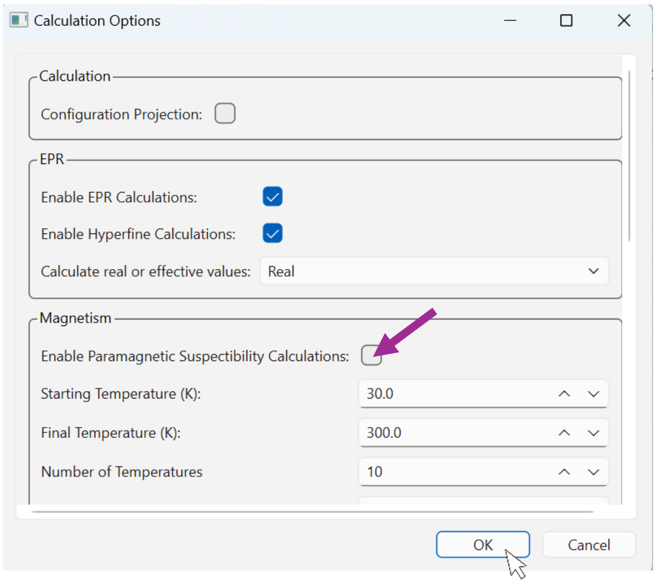
When should EPR parameters be varied manually?

Our advice is to use the estimated parameter values for P and kappa where possible for initial fitting, this will allow the focus to be on fitting *e* parameter values. However, there are some instances where manual tuning will be necessary:

1) If no P value is available for the metal. This is unlikely but for gas-phase calculations with very high or low oxidation states or for 4d/5d transition metals, no tabulated P value will be available so a value will need entering by hand. Complexes with mixed isotopic abundance may also require an adjusted P value.
2) In cases where an excellent fit is obtained to other spectroscopic data, then it is reasonable to adjust kappa slightly to obtain a better fit due to the complexity of its calculation. Whilst KESTREL gives a good approximation, this is only accurate to within ± 10%.

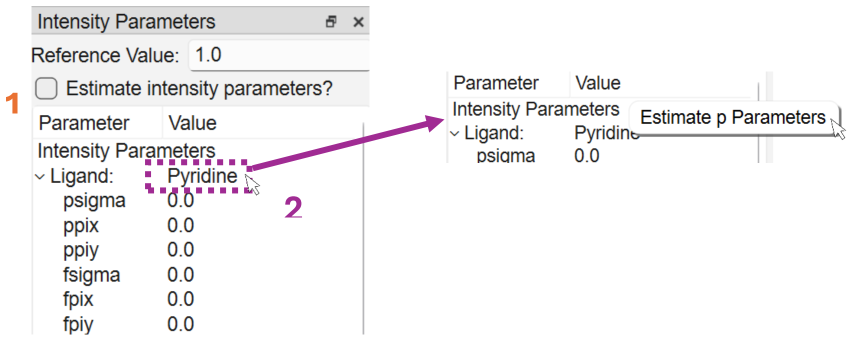
Why are the results of calculation not updating?

This is likely because the calculation is disabled. Try navigating to the 'Calculations' Menu and choosing 'Calculation Options'. This will open a box where you can scroll to the calculation you want to enable. You should now be able to open the relevant window in the 'Calculations' menu to see the results of your calculation.



Why is the UV/MCD/CD spectrum not updating?

Absorption spectra require the input of intensity parameters, otherwise a flat line will be seen even once the calculation has been enabled. These parameters can be estimated by checking the box "Estimate Intensity Parameters" in the "Intensity Parameters" menu.



Glossary

***e* value**	Describe the strength and type of a metal-ligand bond. A larger positive value denotes stronger electron donation from the ligand and a negative value donation from the metal to the ligand (back-bonding). Sigma (σ) values denote σ bonding and pi (π) values π bonding.
Racah B/C	Interelectronic repulsion parameters that describe the strength of repulsion between d electrons. In combination with the ligand field splitting, greater values tend to favour high-spin electronic configurations.
Spin-orbit Coupling (SOC)	Describes the strength of spin-orbit coupling between an electron's spin and its motion around the nucleus. This phenomenon increases along the 3d period as the number of electrons increases. Generally, free-ion values are used to describe the SOC, at least initially, in KESTREL.
Orbital reduction factor (k_{iso})	Describes the quenching of orbital angular momentum due to covalency effects. A value of 1 is generally understood to indicate a perfectly ionic system with decreasing values representing greater covalency.
Ligand Field	This is the premise behind KESTREL's calculations. The d-orbitals are perturbed by the surrounding ligands such that their energies change. The nature of the ligand field surrounding the metal centre is what KESTREL calculates and parameterises. In KESTREL, the ligand field is described using a 5×5 ligand field matrix (see below).
5×5 Ligand Field matrix	This is key to ligand field calculations and, for each ligand, describes its *e* parameters. Ligand field matrices for each ligand are then rotated and summed to give an overall ligand field matrix which denotes the ligand field surrounding the metal centre. Diagonalisation of this matrix gives the energies and composition of the 3d orbitals.
Eigenvalues and eigenvectors	In this context eigenvalue and eigenvector refer to the result from diagonalising a matrix. The eigenvalues describe the principal values, i.e. the d-orbital energies from diagonalising the 5×5 ligand field matrix. The eigenvectors describe the orientation in which a matrix is diagonal, i.e. the d-orbital compositions for the 5×5 ligand field matrix.
Many-electron eigenfunction	Describes an electronic state and its multiplicity. The electron configuration can also be shown. For example, a d^1 system would show ten possible states for a single unpaired electron spin-up or spin-down in 5 d-orbitals.
Zero-field splitting (ZFS)	Describes the splitting of energy levels, due to spin-orbit coupling (SOC) that exists in the absence of a magnetic field. For systems with more than one unpaired electron, this causes the ground state to split. For systems with an odd number of unpaired electrons, this splitting results in the formation of Kramer's doublets (each containing two electronic states). In cases where this splitting is large, it is often assumed that only the ground multiplet is occupied, leading to an effective S=1/2 system.

References

1. M. Gerloch, *Magnetism and ligand-field analysis*, Cambridge University Press, Cambridge, 1st Edition edn., 1983.
2. L. J. Kershaw Cook, R. Kulmaczewski, R. Mohammed, S. Dudley, S. A. Barrett, M. A. Little, R. J. Deeth and M. A. Halcrow, *Angew Chem Int Ed Engl*, 2016, **55**, 4327-4331.
3. S. Stoll and A. Schweiger, *J Magn Reson*, 2006, **178**, 42-55.
4. D. C. F. E. Mabbs, *Electron Paramagnetic Resonance of d Transition Metal Complexes*, Elsevier Science Publishers, Amsterdam, 1992.
5. A. K. K. A. D. J. Miller, *Atomic Data and Nuclear Data Tables*, 1985, **33**.
6. P. K. a. M. K. Marketa L. Munzarova, *J Am Chem Soc*, 2000, **122**, 11900-11913.
7. P. Comba, G. Nunn, F. Scherz and P. H. Walton, *Faraday Discussions*, 2022, **234**, 232-244.
8. G. V.-K. Svein Jarle Horn, Bjørge Westereng and Vincent GH Eijsink, *Biotechnology for Biofuels* 2012, **5**.

Citation Information

If you use KESTREL in your work, we kindly ask that you provide the following citation:

- G. Nunn, B. S. Grace, M. A. Bates and P. H. Walton, KESTREL [Computer software]. https://github.com/chem-kestrel/kestrelpy

License Information